

The Concept of Analogy

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Introduction

1. In this treatise I have set myself the task of continuing the closer investigation of [3] some of the fundamental concepts which, in my book *Den menneskelige Tanke* [The Human Thought] (1910), I attempted to exhibit as the most characteristic for human cognition, such as we know it after its development hitherto. I have thus wished to give a companion piece to the treatise on *Totalitet som Kategori* (Videnskabernes Selskabs Skrifter. 1917) and to the treatise on *Relation som Kategori* (Videnskabernes Selskabs filosofiske Meddelelser. 1920), and it is the concept of analogy that I shall here investigate more closely.

Provisionally, analogy can be defined as likeness of relation between two subject-matters, that is to say, a likeness which does not hold of single properties of those subject-matters or of single parts of them, but of the relation of properties to one another or of parts to one another. There may be a very great difference between two subject-matters' properties or parts, each taken by itself, and yet there may be a likeness in the relation between the properties or between the parts. Analogy is thus a peculiar kind of likeness. It stands in a certain sense further from identity (absolute likeness) than qualitative likeness, composite likeness and coincidence do, which concern only single properties or parts, and which more easily form transitional forms to identity (cf. *Den menneskelige Tanke* § 21). Yet it must be remarked that there is not only an identity of subject-matter, but also an identity of relation, and already Aristotle distinguishes [4] between quantitative analogy (ἰσότης λόγων), proportionality in the strict sense, and qualitative analogy, which may hold, e.g., between geometrical figures or between organs in different living beings. And there is besides an identity of difference, which lies at the base of the formation of an identically varying difference-series (*Den menneskelige Tanke* § 65) and especially of the formation of the identical quality-series that are fundamental

for exact science (number, time, degree, place) (*Den menneskelige Tanke* §§ 76–82).

Both totality and relation I have given a place among the categories, that is to say, among the fundamental concepts, the forms of thought, which at science's present stage play an essential role in thought-work, determining its tasks and its whole direction (*Den menneskelige Tanke* §§ 51–52; 57). Neither Aristotle nor Kant has given analogy a place among the categories, and in *Den menneskelige Tanke* I followed them in this. Yet in that work I was led to lay much weight on the fact that, in contrast to purely formal thinking (logic and arithmetic), analogy plays an essential role in all thinking about real subject-matters. The opposition between identity (identity of subject-matter in logic, identity of difference and identity of relation in arithmetic) and analogy is therefore of great importance for the right apprehension of the nature and the validity of our cognition. But when analogy really plays so great a role, while identity is properly an unattainable ideal outside the formal world of abstractions, it might be warranted to give it a place in the series of the categories; and in the table of categories to be found on the last page of the treatise *Totalitet som Kategori* this has in fact been done. It is entered there among the formal categories, immediately after the concept of identity, [5] because it lies at the base of that reduction of relations of quality which — with the help of points of view of identity — takes place in the stricter development of the concepts of number, time, degree and place. Such series form intermediate members between absolute identity-series and qualitatively different series of subject-matters. They are formed by analogizing abstraction from the latter. It is here that analogy especially shows its importance. It should now be the task of the present treatise to follow the concept of analogy further in its more special importance for human cognition.

I dwell first on the involuntary analogies which make themselves felt in primitive men and in children, but also, at the high points of the life of the spirit, in the genius. They do not come forward as analogies for the consciousness in question itself; that is possible only for the critical consciousness, which has learned to distinguish between analogy and identity and to see that every grounding and proving transition of thought takes place in virtue of identity. Thereafter I go on to investigate analogy from a formal logical point of view, and in the following sections I seek to exhibit analogies between cognition's different functions and between its different domains.

Provisionally it may be remarked that analogy's importance for cognition shows itself

in three different respects.

Its greatest importance it certainly has by working in a scouting, anticipating and spurring manner. Categories in general lay themselves open to view in the questions that are put (*Den menneskelige Tanke* § 52). Analogy may be the actual road to true cognition, even if the analogy's validity is afterwards rejected. One can, after all, draw a correct conclusion from false premisses. Naturally one cannot prove that the conclusion is correct except by finding the right premisses; but it often happens that an assumption [6] stands for us as true, and that precisely thereby it sets us the task of finding out why it is at bottom true. The premisses (the right premisses) are often the last thing to be found; but one would not seek after them if an analogy had not first led us to an assumption and thereby set us a task. The analytic method rests on this, that a task has been set us, and that one then seeks the conditions for solving that task. If the right premisses are then found, one perhaps rejects the analogy that showed the way — not only if it is false, but also if it is unnecessary. The ground for an analogy's being set up will lie in this, that there is a break in a continuous connection which has formed itself involuntarily for us, and which an involuntary need leads us to hold fast as long as possible. Analogy may here lead to an intellectual venture, which is sometimes crowned with success. One may apply to analogy what Pascal says of the imagination in general, that it deceives all the more easily because it does not always deceive (*d'autant plus fourbe qu'elle ne l'est pas toujours*). If it does not deceive, or if it shows itself to be dispensable, it is replaced by a determination of relation which is finally grounded in identity.

But analogy is not merely preparatory. It may be the last bond of union between different subject-matters or series of subject-matters which cannot be reduced in such a way that a single subject-matter or a single series of subject-matters is seen to lie at the base, or in such a way that they are all seen to have their ground in more deeply lying subject-matters or series of subject-matters. A certain rounding-off and survey becomes possible, in the face of such subject-matters, only by the help of analogy. We do not thereby obtain an explanation of why there are different subject-matters or series of subject-matters, but we establish the actual state of the case, namely that at certain points [7] we get no further than to a connection, a synthesis, which builds only on the remotest degree of likeness — which analogy is. It will appear that both in the face of cognition's different functions (see ch. III) and in the face of its different domains (see ch. IV) we are

finally placed in this way. We end then in open questions, since it remains undecided whether a further reduction, a more deeply lying continuity, might be found, or whether there are here, in the nature of things, differences which are at any rate irreducible for us. Questions can thus be not only the beginning of wisdom, but also its end.

It will have its interest, finally, to dwell somewhat on analogies whose importance lies in their being expressions of poetic and religious moods — expressions which may recall now the analogies of involuntary mental life, now the seeking and finding of scientific thinking. Especially will intimations or hopes of a unity and a continuity lying deeper than experience can exhibit find expression in this kind of analogy.

Chapter I

Involuntary Analogies

2. As already remarked, analogy does not always presuppose that a distinction has expressly been drawn between different subject-matters, or respectively between their properties or parts, and that afterwards, in a certain opposition to the differences, a connection is brought about between the subject-matters on the basis of a likeness of relation. In involuntary mental phenomena, such as sense-intuition, memory and imagination, it cannot always be shown that the elements are first present as separated and only afterwards enter into combination. The importance of these involuntary mental processes lies precisely in this, that they bring about wholes, combinations, which can be analysed only afterwards. Reflection constantly needs a material, but this material need not, as has often been believed, consist in a multitude of independent elements; it may be wholes, which it is then reflection's task to resolve into elements that may not without more ado be regarded as identical with the elements to which the involuntarily formed wholes are due. For newly emerging elements may at once and involuntarily be woven together with consciousness's given content, so that perhaps no alteration at all is noticed. All the same, there is a tendency in reflection to regard the elements which its analysis finds as the whole's original, elementary foundation. There is here an analogy present, which at most can seek its justification in the continuity with which the transition can take place from an involuntarily formed mental continuum to the differentiation to which the very arising of reflection already testifies. Properly, when we speak of the most primitive and involuntary mental processes, all designations fall short, and this is no wonder, since language's psychological designations are fetched from the [8]

clearly differentiating, critically reflecting mental life. The very expression “analogy” we here use in an analogizing manner. It is here as when one speaks of “unconscious inferences”. These are not inferences in the proper sense, but processes, transitions, which can be interpreted in analogy with the proper inferences. The same holds further (as I have already shown in my *Psykologi* and later in *Den menneskelige Tanke* § 5) when we set up the concept of synthesis as an expression for the nature of all conscious life. In epistemology, where we investigate the possibilities of scientific cognition, we must also [9] discuss the possibility that the given might consist of a heap of equally great and mutually irreducible differences. Only an approximation to such a chaotic difference-series can be exhibited (as also to its counterpart, an absolute identity-series); there would in any case be needed a searching and strenuous reflection to establish that such a series really lay before us in a determinate given case.

In the manner in which a single lived experience, from the very moment of its conception, is connected with other simultaneous or past lived experiences, or with traditions into which the individual has lived himself so that these too can be regarded as lived experiences, there lies already an interpretation. The further back we come towards what we call the primitive man’s stage, the closer lived experience and interpretation move to one another, and finally there is no opposition between them to be found. When we nevertheless attempt to hold them apart, we make use of an analogy with what first makes itself felt for testing reflection. And, as has been said, the very concept of analogy is used analogically when we say that primitive men interpret every new lived experience in analogy with what they know from the traditions into which they have lived themselves. — In my treatise *Oplevelse og Tydning* I have sought to describe different ways in which the relation between lived experience and interpretation comes forward historically, as regards ecstatic experiences. —

In his important work *Les fonctions mentales dans les sociétés inférieures* (1910), LÉVY-BRUHL has criticized the formerly common conception that primitive man apprehends everything on the likeness of himself, anthropomorphizes or analogizes. It is, on his conception, not comparison or reflection that is at work here, but an entirely immediate [10] identifying. In primitive man there always lie traditions at the base, with which the new that is lived through involuntarily melts together. The representations handed down from the tribe make themselves felt with such strength of feeling that no need

for reflection arises. What deviates in high degree from what has been handed down is provisionally not noticed; only later is an interpretation perhaps thought out that agrees with the tradition. When the savage sees an image of an object he knows, the image as image brings him nothing new, for the image is the object itself, however he may afterwards arrange this relation for himself. And with all the logical consistency which already distinguishes this stage — which Lévy-Bruhl has wrongly called the “prelogical” — the savage therefore fears to become dependent on one who has an image of him, and he fears that he will come to suffer want of game when a European has drawn or photographed many wild animals in the district and has departed with the pictures. And as it goes with images, so it goes also with names and with ceremonies: they are what they mean. Images, names and ceremonies are indeed only a part of a whole which has other properties besides those that are depicted or named; but this part *is* the whole. Lévy-Bruhl finds in this identifying of part and whole — what he calls “loi de participation”¹ — the most characteristic peculiarity of the primitive mental life. It is certain that the interpretation which primitive man institutes under the influence of a tradition embraced with heartfelt adherence and reverence cannot follow the principles of formal logic. These make themselves felt only when doubt has been awakened; but he is accustomed to regard the tradition as a self-evident *foundation* for his apprehension of everything that happens. The basis of interpretation is not the work of reflection. But [11] it may lead to misunderstanding when Lévy-Bruhl declares (p. 425) that the collective representations governed by the “law of participation” are “indifferent to the logical law of contradiction”. A distinction must be drawn between the given basis of interpretation (here: the tradition held fast in reverence, which is once for all stamped by the “law of participation”) and the interpretation of single phenomena that takes place on the ground of this basis. Reflection works at all times and on all levels of culture on the ground of determinate presuppositions which cannot themselves at once be made an object for reflection. On the basis which primitive man once for all has, it is entirely consistent that he does not understand why the European distinguishes between the image and the thing. When Lévy-Bruhl calls “the primitive mentality” “prelogical”, this fits the basis of interpretation, not the interpretation made on the ground of that basis.² As there will

¹To this corresponds what GRØNBECH (*Primitiv Religion*, p. 8) calls “Helhedsrealisme” [realism of the whole], and which is characteristic for the primitive mental life.

²Dr. ALBERT SCHWEITZER, who works as a physician in the primeval forests of equatorial Africa, writes,

later be occasion to mention, Lévy-Bruhl has himself in another connection maintained, and rightly, that human consciousness, as regards the ultimate basis for judgments and inferences, properly never quite loses its “prelogical” character. There will always be presuppositions present which we do not and cannot bring forward, even in our most logical reflection. — As regards especially the close connection between name and thing, one can find examples of it even today. There is a tendency to regard the terminology one uses oneself as the only “right” one, and the language one speaks oneself seems to [12] be one with the thing, while foreign languages’ designations often seem foreign to the things. It is told of a German who visited Paris that he reported home that there they called bread “pain”, and added, “Wir nennen es Brot, und es *ist* doch auch Brot!”

Tradition comes forward for the savage as a closed whole, within which he has no ground for making any distinction. There provisionally rules an immediate belief in the validity of the representations handed down, just as in the sense-perceptions that immediately appear. And the expression “belief in” is even too reflected, for no distinction at all is drawn between the belief and that which is believed in. The very concept of reality or validity is first formed when doubt has made itself felt. The fundamental proposition in accordance with which thinking provisionally goes (though it is of course not formulated) can be expressed by the question “Why not?” Why not rely on everything that comes forward for us with living force as sensation or as tradition? Only later does one learn to ask “Why?” — and reflection can then even reach the point of directing this question against the very thing that has hitherto formed the undisputed basis. For that it is required that new lived experiences should appear so numerous, and with such a peculiarity deviating from everything customary, that they can gain power over what one has already lived oneself into. ALBERT SCHWEITZER declares, as the result of several years’ living together with negroes in Central Africa: “One must not believe that one has got to the bottom of the negro’s world of thought if one has got no further than to look at his superstitious representations and his inherited rules of law. Innermost in him there lives a dark intimation that the apprehension of what is good must come by way of reflection” (*Mellem Floder og Urskove* p. 136). The self-sacrificing physician, who through [13] historical criticism had learned to distinguish between the Christian dogmas and the

after several years’ stay among the negroes: “The natural man thinks more deeply than one commonly believes. Even if he can neither read nor write, he yet institutes reflections to a far wider extent than we suspect.” (*Mellem Floder og Urskov*. Danish transl. 1922. p. 134.)

ethical idealism that comes forward in the Sermon on the Mount and in the parables, experienced this especially when he sought to explain his religious apprehension to the negroes. Yet he experienced also that, alongside the “morality of the heart” that could be called forth in them, there long maintained itself a desire to lie and to steal, and a need for revenge which according to their own traditions was their right and perhaps even their duty.

I must here interpolate the remark that when in my *Psykologiske Undersøgelser* (p. 24 f.) I say that in the relation between new lived experiences and practised, customary representations the chief weight lies on the new, which “sets the whole process going”, this presupposes another stage of development than the undisturbed primitive mentality, where the new so easily glances off against the customary and is able to gain power in the mind only when it comes forward in a particularly insistent manner. —

If one asks after the origin of the system of collective representations that makes itself felt in the savage and determines his interpretation of what meets him, Lévy-Bruhl answers that it “is lost in the night of time”. And since every human group hitherto met with comes forward with certain collective representations, we cannot exhibit any absolutely first beginning. We must here, as with the question of the origin of language,³ content ourselves with analogies fetched from the levels of culture that lie nearest to us, and these analogies must of course be used with great caution. We see that on the [14] levels of culture we know, the traditions that lie at the base of each new generation’s spiritual development undergo constant alterations, additions and displacements, which for an essential part are due precisely to the new generation’s own experiences and reflection. The collective is constantly altered under the influence of the individual. We cannot, as was often attempted in earlier times, think of a time when there was nothing collective, or when what collective there was was due exclusively to single individuals’ experience and reflection. But just as little can we think of a time when minds were governed exclusively by common traditions, without individual experiences and thoughts exercising any influence whatever. It must be assumed that on all levels of development there has taken place a reciprocal action between the common and the individual, and

³Cf. OTTO JESPERSEN: *Language, its Nature, Development and Origin*. (1922). p. 416 f. on the method in the discussion of the question of the origin of language. One must (as already, in their way, Madvig and Whitney had maintained) start out from the manner in which language actually develops, that is, from its history, and from there infer back to a more primitive language than any of those we know.

the difference of the levels can consist only in this, that on earlier levels the common, on later levels the individual, has the greatest influence — yet so that there is again and again a swinging to and fro between the different factors' predominance. Even on the most primitive levels we know, individual differences can make themselves felt and deviations from the tribe's belief come forward.⁴ And the examples of such things would undoubtedly be still more numerous if we had more knowledge of life within primitive societies than we have. We should have found remarkable deviations and initiatives which now are drowned for us in the ruling traditions.

Primitive conditions of life present great uniformity, and therewith great community, in the experiences which the single individuals can make. And on account of the constant [15] living together in group and family, the single man's special experiences and inventions will very quickly become common property, and his name will be forgotten. Added to this, the practical demand constantly makes itself felt; action is required, and life is for a great part lived as reflex-life and instinct-life. Every representation gets its essential importance by releasing movement. But for that it is not necessary to enter into the whole connection to which the representation's content belongs. A single part or a single property is enough to release action, and one need not distinguish between part and whole; the part stands involuntarily as a sign of the whole. The image, the name or the sign works as if the object itself were given in it. The instinct has enough in an intimation to begin to stir. In the fact that a part can thus have the same influence as the whole, and that it may even be advantageous that time and force are not expended on investigating whether the whole is really represented by the single part or property, there might lie an access to an understanding of the "law of participation" or the "realism of the whole". And when a part thus shows itself again and again, in the race's experience, to have the same practical importance as the whole, one sees the possibility that there may be formed a circle of collective representations which obey the law of participation, and which it may be difficult for new lived experiences to break.

Purely logically, this identifying of part and whole (*pars pro toto*) can be formulated as an inference in the second figure: the part releases a certain movement, which the whole would release; therefore part and whole are the same! And this is precisely one of the ways in which an inference from analogy can be formulated. When an instinctive

⁴See my treatise *Sociologi og Religionsfilosofi* (in the book *Oplevelse og Tydning*. 1918) p. 131–138.

action can lead astray, this hangs together with the fact that the “inference” in question is wrong. Only within quite determinate limits, which the consciousness governed by an unconditional impulse to action does not give itself time to establish, can it do to identify [16] part and whole. Neither instinct nor prelogical mentality can therefore maintain itself longer than the inference from analogy practically suffices. Doubt will be awakened when this is no longer the case. A closer psychological investigation will be able to elucidate this relation more nearly and thereby cast light on the involuntary analogies.

3. Just as the adult European is too inclined to read his own developed and reflecting mental life into the mental life of the so-called primitive men, so there is in general a tendency to read functions that presuppose a certain development of reflection into the most immediate and involuntary mental processes. One then speaks of association of representations and of judging, where in reality it is more elementary processes that are going on. This comes in part of the fact that language’s expressions for mental life are essentially fetched from the developed, differentiated mental life, and one is then not clear that it can be warranted only by way of analogy to use such expressions of the most elementary psychical processes. Some years ago (*Psykologiske Undersøgelser*. Vid. Selsk. Skr. 1889) I had occasion to draw attention to this, and I now bring forward a few single points from that investigation, especially in order to show that what Lévy-Bruhl has exhibited for the mental life of primitive men is found again in mental processes that go on in all of us to this day.

a. It has not been sufficiently noticed that there are several different kinds or forms of recognition. In the treatise just mentioned (p. 35 f.) I thought I had to distinguish at least ten kinds of recognition. It leads to confusion when one speaks, as is commonly [17] done, of recognition quite in general and yet in fact keeps to a single form — and that a form which presupposes a certain differentiated development of the life of representation. There is then a failure to recognize the peculiarity of the elementary, involuntary mental life.

The most elementary form of recognition that can be exhibited is as immediate and as involuntary as a sense-sensation can be. A subject-matter here stands for us at one stroke as familiar, with a quality of familiarity, just as a sensation has its sense-quality, without any association, comparison or judgment being demonstrable. A name, a facial expression, a shade of colour can appear with a quality of familiarity without our being

able to remember that we have observed the same subject-matter before, and without our being able to refer it to any determinate connection — that is, without knowing who bears the name, and so on. Perhaps the quality of familiarity gives us occasion to form the judgment that we have heard the name before, and so on. But that judgment then comes precisely afterwards. And if one will conceive such immediate recognition as association, one must formulate it as a limiting case and think of the matter thus: that when a sensation of sight or of sound is repeated, it can, in the very moment of its birth, fuse with the after-effect of earlier sensations of the same character (however one may care to conceive such an after-effect). There takes place, then, a momentary reproduction, and what is reproduced fuses at once with the sensation that is coming into being. The quality of familiarity can then theoretically be conceived as the product of two elements. Physiologically regarded, we have here an example of practice. All practice of a function [18] is due to a fusion of a new release with the after-effects of earlier releases.

This exhibiting of immediate recognition as an elementary phenomenon, lying on the boundary between sensation and memory proper (I have called it bound memory), was not acknowledged or attended to when it first appeared. It was rejected by Danish psychologists, and writers such as Wundt, Meinong and Bergson assumed without more ado that association, comparison or judgment is always the condition of recognition. It has interested me the more to see that G. E. MÜLLER, who is an authority as regards the doctrine of representations, in his psychological lectures printed as manuscript (1913) § 26 describes the same phenomenon that I call immediate recognition, under the name of indeterminate recognition. “Recognition of an object of observation is,” he says, “*indeterminate* when, in observing the object, we merely have the impression that we must somewhere have observed it before. When the object not only comes forward for us as familiar, but in addition awakens the recollection of its name, of the circumstances under which we observed it earlier, and so on, then we have a *determinate* recognition.” I have against this description only this to object, that the quality of familiarity *need not* be bound up with a consciousness of (“the impression of”) our having apprehended the object before. It *can* be without consequence. It is already an inference when we take it that we have apprehended the object before — an inference drawn from the difference between sensations with and without the quality of familiarity, an immediate inference of course. When immediate recognition at once releases a reflex or an instinctive action,

no such inference is drawn, and “the impression” will not make itself felt.

MÜLLER finds (in a handwritten addition to the printed lecture) the explanation of [19] indeterminate recognition in this, that after-effects of earlier observations, and perhaps indistinct reproductions of these, are released by the new sensation; or that subject-matters we have apprehended earlier are, on repetition, apprehended more easily and more quickly than before. In quite the same way I had also referred to the law of practice, which finds application both to the psychological and to the physiological side of the phenomenon. It can of course often be difficult to decide whether, and in what degree, an indistinct association of representations takes place. I have therefore designated immediate recognition as a limiting case, and I have in my *Psychologi* compared the relation between immediate recognition and association of representations with the relation between the tangent and the secant: the nearer the secant’s two points of intersection with the circle come to lie to one another, the more does the secant approach the tangent. Repeated and exact observation has shown me many degrees here, and there is every occasion for closer investigation.

Even before MÜLLER’s lectures were printed, BETZ (*Archiv für die gesamte Psychologie*. 1910, p. 267) had declared: “Erst die vielzitierte Abhandlung von Høffding über Wiedererkennen zeigte, dass die Dinge bei weitem nicht so einfach liegen, als es den Assoziationstheoretikern schien”. — The Neoplatonist Olympiodorus,⁵ moreover, had already mentioned the phenomenon. In his commentary on Plato’s *Phaedo* he supplements his master’s doctrine of recollection by drawing attention to the fact that there is not only a transition from one element to another on account of likeness [association by likeness], or from one element to another element different from it with which it [20] has earlier occurred together [association by contiguity], but that also “the same can be connected with the same”. When someone recognizes Socrates, for instance, after having forgotten him for a time, then it is, says Olympiodorus, just as much a sensation as the first time he saw Socrates; in such recognition there takes place no transition of cognition (οὐ γέγονε μετάβασις γνώσεως). That Plato does not mention this kind of recognition, Olympiodorus explains by the fact that in the *Phaedo* he had use only for the free (“determinate”) recognition, in which, according to him, “the Ideas sprang forth in the soul”.

⁵ *Olympiodori in Phædonem Commentaria*, ed. W. NORVIN. 1913. p. 68.

b. When there lies before us a composite subject-matter whose elements (parts or properties) we have often apprehended in succession, we need not always go through all the elements anew for recognition to take place. Successive recognition can be replaced by partial recognition. If the subject-matter's elements are $a b c d$, it can be enough that a is recognized. We have here quite the same phenomenon that LÉVY-BRUHL has designated as participation. Partial recognition has great practical importance, since it would often take too long a time if the whole series of elements had to be recognized before we were sure of having the subject-matter before us, so that we could proceed to act towards it. The subject-matter thus contains more than we notice, but we confine ourselves to the strictly necessary. We recognize, e.g., a man by his walk or by some peculiarity of his dress, a chemical substance by its smell, and so on. As with immediate, indeterminate recognition, it is the need for action that shortens the process. The necessity of intervening, if the recognition holds good, weighs practically more than the certainty that the apprehension [21] is correct. When spontaneous or instinctive action leads into misfortune, it is because the single mark does not hold good.

Partial recognition can become sense-illusion when the immediately recognized element has in different cases been connected with different other elements. When both the series $a b c d$ and the series $a f g h$ have occurred, and I, when a comes again, at once rely on $b c d$ being there too, I can be surprised by $f g h$ presenting themselves instead. I use an example from my *Psykologi*. On a walk along the Jammerbugt on Jutland's west coast it happened to me, a whole day through, that I took pieces of wreckage sticking up out of the sand for men. There took place here an involuntary supplementing of what was really seen. The image that involuntarily formed itself for me contained more than proved to be given. The supplementing here takes place just as involuntarily as in the well-known experience with the blind spot, or as when, after a part of the lower lip has been locally anaesthetized, one feels a hole or a depression in the rim of the glass one drinks from. Such a supplementing lies, seen from our modern point of view, at the base of primitive man's participation. The "prelogical", which only by a questionable analogy can be described as having arisen through association or through judging, we thus know from daily experience. Purely logically, sense-illusion can be formulated as a fallacy; but it is all the same unwarranted on that account to read such an inference, as an actually occurring psychical process, into the elementary phenomena we here have to do

with. We simply have no right to identify the elements which analysing reflection finds with those that co-operate in the original lived experience. Here the analogy between psychological analysis and chemical analysis fails.

When one formulates the sense-illusion, or partial recognition in general, as an inference, it is the second or the third figure that we get. The property *B* is found both in the subject-matter *A* and in the subject-matter *C*; from this is inferred partial or absolute identity of *A* and *C*. (Second figure.) Or: the subject-matter *B* has both the property *A* and the property *C*; from this is inferred that there is partial or absolute identity between *A* and *C*. (Third figure.) As we shall see, we have here two of the ways in which an inference from analogy can logically be formulated. What is questionable in every inference from analogy already comes forward clearly here. — [22]

In all participation, that is, in immediate, partial and illusory recognition, an involuntary need for whole-formation makes itself felt. Even if only a single element is actually given, one at once makes it into a whole. GRØNBECH has therefore rightly designated what Lévy-Bruhl calls participation as realism of the whole. Such an involuntary impulse toward the whole is especially characteristic of the first stages of development of conscious life, which bear the stamp of naivety; it does not — fortunately — disappear later either, though it becomes an object of analysis and criticism. It lies at the base of what I have called concrete and practical intuition. By concrete intuition I understand sense-intuition, memory or imagination. In all these three forms, which do not from the first stand as different in principle from one another, a whole is built up on the occasion of a single element or of a few elements. Since it will often be a ruling feeling that determines the centre about which the whole-formation takes place, concrete intuition is more or less akin to practical intuition, in which all the mind's forces can work along, without there needing to be any consciousness of the sources of the whole-image or of the energy with which it is formed and held fast. All practical faith, indeed all personal concentration upon a thought or a task, is of this kind. Here too we are reminded of the "prelogical" consciousness, which according to Lévy-Bruhl has a purely emotional and mystical character, since it is the tribe's sacred traditions that decide how the individual interprets, or rather integrates, what he lives through.⁶ [23]

⁶Cf. my treatise on *Begrebet Intuition* (Vid. Selsk. Oversigt 1914) p. 75–77 and my work *Henri Bergson's Filosofi* p. 22 f. — In contrast to the two kinds of intuition here mentioned I set analytic and synthetic intuition; the former expresses reflection's testing, the latter its gathering function.

c. In all recognition, likeness to what has earlier been lived through plays a role, and to that extent recognition can be regarded as a limiting case of association by likeness. As regards association by likeness in general, there has been dispute whether any such thing exists at all. My position on this question has, ever since the first edition of my *Psykologi*, been that I do not assume any pure and unmixed association by likeness, since in all demonstrable examples of likeness there will always be certain differences present. No one has been clearer about this than Plato — it is of course for that reason that he makes perfect likeness an “Idea”. And the elements of difference will be able to release association by contiguity, which plays a greater or a smaller role in the transition to the following representation. Every actual association will therefore be a mixture of association by likeness and association by contiguity, and the discussion must in each single case turn on the degree to which moments of likeness co-operate. My formula for association by likeness is therefore $a_1 + a_2$, and the indices designate the differences which always make themselves felt, even where the likeness is preponderant. Since it is peculiar to such transitions in the life of representation as are essentially due to likeness [24] that they take place involuntarily, and especially when the mind is strongly occupied and moved, it will be difficult to exhibit such transitions experimentally; it is very difficult in psychological experiments to avoid preparation and expectation, whereby association by contiguity is at once favoured.⁷

While many psychologists (especially Danish ones) took a quite dismissive attitude towards the assumption of an association by likeness, even with the limitations stated, G. E. MÜLLER, who has quite especially investigated connections of representations, expressed himself more cautiously. He did not find it probable (in the lectures mentioned above) that there was pure association by likeness; when a portrait calls forth the representation of the person himself, it may happen by its first calling forth the name, and this then the person. There is of course no doubt that it can go this way. I have, however, not been able to convince myself that it fits all cases; I can well recognize a portrait even if I have forgotten the name, which perhaps even after the recognition I have trouble in recalling.

In the same year in which Müller had his lectures printed as manuscript, he took part as a subject in some experiments conducted by his pupil Louise Schlüter, which were to test the relation between two methods of teaching a foreign language, of which the

⁷See, besides my *Psykologi: Psykologiske Undersøgelser* (1889), p. 43–59.

one builds on a direct connection between an object and the foreign word (the intuition method), while the other first names the domestic name of the object and lets that recall the foreign word. It showed itself here that when, the day after an object had been shown and the foreign word learned, a similar object was shown, the foreign word came up [25] without any support from the domestic word. This presupposes that the object seen on the preceding day was reproduced from the object now seen, although it was not quite the same. This Louise Schlüter regards as proof that there can be a real association by likeness,⁸ and Müller has declared himself in agreement.⁹

I shall only remark on this, that in observations and experiments on this relation a distinction must be drawn between the different kinds of likeness. In Louise Schlüter's example it was, so far as I can see, qualitative likeness that was in question: the object shown on the second day was not the same as the one seen on the first day, but "ein ähnliches Exemplar derselben Art". It was thus not what I call coincidence that made itself felt here. But since the relation between examples of the same species is for the most part likeness of relation — inasmuch as it need not be a likeness with respect to a single part or property that is decisive for the species-concept — it would be worth investigating how, in cases like Schlüter's, qualitative likeness and likeness of relation stand to one another.

At bottom, Schlüter's case is not different from the cases on which MÜLLER in his lectures (§ 20) has built the form of association which he calls active substitution, and which is an association by likeness based on an association by contiguity. When a is associated with the representation b different from it, then a representation α which resembles a can also call forth b , whether the relation between a and α is qualitative likeness or analogy.

Such active substitution is of great interest for our investigation of analogy in its [26] epistemological respect. For when it shows itself that a and α practically play the same role, whatever kind of likeness it may be, we see here a psychological possibility for that confusion of identity and analogy which proves so fateful in the history of thought. There will from the first be no doubt about the validity of the transition from α to b . Practically

⁸*Beiträge zur Prüfung der Anschauungs- und der Übersetzungsmethoden* (Zeitschrift für Psychologie. 1913) p. 73; 103.

⁹In handwritten additions to a copy of the lectures, which Professor Axel Herrlin has kindly placed at my disposal.

regarded, there is a tendency to treat all likeness as identity. This is seen already from the much-discussed phenomenon that a burnt child dreads the fire. According to Stuart Mill, we have here merely an association by contiguity. But the child provisionally shuns all objects that resemble the one on which it once burnt itself.

4. In the child's play and language, in artistic intuiting and in the genius's working, involuntary analogies make themselves felt which are as little as those discussed in the foregoing due to express comparison or inference, and are not due either to an association in which every single member is apprehended by itself and only thereafter connected with other members.

a. The child's play is often due to imitation of the grown-ups' doings, but the child can immerse itself in it with so complete a living-into that the difference between what the play "is" and what it is to "mean" does not make itself felt at all. If the play is interrupted from outside, this can awaken pain and indignation. A child was playing housewife and had laid its play-table with various imagined dishes. When it now became the house's dinner-time, and the child was wanted to sit down at the real table with the rest of the family, it became violently incensed and burst into tears, "for a mother doesn't just run away like that from the table she has to look after". The indignation shows how deep the living-into the play had been. In a similar way it can go with the spectator at a drama, [27] when he is suddenly disturbed in his full merging into what he sees and hears. It has wrongly been attempted to explain the attraction of a tragic drama by supposing that the reader or spectator is conscious of standing outside what is presented. Such a knowledge would make the full absorption impossible. The absorption presupposes that we for a while identify our own fate with the one presented. It is an illusion, a "participation", which is perhaps lifted when the curtain falls or the book ends, but whose resonance or "perseveration" can make itself felt for a long time, since we may have a tendency to live again and again in the same mood as when we were immediately under the drama's influence.

Outside play, too, children show a capacity for analogical lived experiences. A state or a situation that has brought them satisfaction they will have continued, either in such a way that they themselves are still part of it, or, more disinterestedly, so that the state is merely continued after the same scheme as before. In both cases it is equivalents that are sought. AVENARIUS has, in his interesting principal work *Kritik der reinen Erfahrung*,

sought to show that here — in the need to continue a state, even if one thereby goes beyond the old forms — a universally human tendency makes itself felt. He adduces the following amusing example from child psychology (which decidedly does not belong among the disinterested ones). A little boy is sitting comfortably enough in an armchair and letting his guest, a little girl, stand. His mother reproaches him for it, and he vacates the seat, whereupon the girl settles herself in the chair with her toy on her lap. The boy looks at her a little and then proposes that she should lay the toy on the floor and let him sit on her lap.¹⁰ — To the disinterested continuation of a situation, on the other hand, [28] belongs the following example. A little boy was riding on his grandfather's knee, but had to break off the situation when he was to go to bed. A little after, he came rushing back with great eagerness, bringing his toy bear, which he placed on the grandfather's knee. In this way a new member was set into the relation, which otherwise would have had to cease. Perhaps there was also sympathy with the old man, whose lap was now empty. —

In the child's acquisition of language, not only recollection and imitation of the grown-ups' speech, but also involuntary analogizing, plays a role. The child has, e.g., heard the expression "Jacob's hat" together with the corresponding visual image, and now forms on its own account the expression "Peter's hat" on a given occasion, without the general representation "hat" being a necessary intermediate member. Children moreover often form regular verbal forms on their own account, when they have not learned the irregular ones. Afterwards they can then practise using the irregular forms which usage requires, instead of the regular ones they had themselves formed in involuntary analogizing.¹¹

b. The fact that creation by genius was comparatively late acknowledged as something other and more than the product of a mechanical joining-together of particulars in a conscious purpose¹² shows how hard it has been to grasp involuntary psychical whole-formation. All mental processes were regarded from the standpoint of analysing reflection. [29] After Rousseau (especially in the article "Génie" in his *Dictionnaire de Musique*) had described genius as the capacity for involuntary creation, and had entreated "l'homme

¹⁰*Kritik der reinen Erfahrung*² II p. 493, cf. I, p. 136. — AVENARIUS finds the same tendency in such cases as those where men, through sudden blows of fate or through the enfeeblement of age, can continue their spiritual life only by the help of forms of faith which in their vigorous years they had abandoned. *Kritik der reinen Erfahrung*² II, p. 295.

¹¹OTTO JESPERSEN: *Language*, p. 130 f.

¹²Cf., e.g., the way in which HOLBERG himself explains the coming into being of his various works. See my *Mindre Arbejder* II, p. 150–154.

vulgaire” not to misuse this sublime word, it was KANT who fixed the concept in the form in which it is now commonly used, in apprehending genius as that rare originality of natural endowment which makes it possible to bring forth works of exemplary character — works which cannot be imitated, but which can awaken originality in other natural endowments (*Kritik der Urtheilskraft* § 40). Genius is, like instinct, an immediate union of need and capacity. The involuntary and the exemplary together make up the essence of genius. GOETHE rejoiced (*Dichtung und Wahrheit*. 19. Buch) that the best word for involuntary and model-worthy creation was thus rescued. — A fate like that of the word genius threatens the word imagination, when one uses it of all free representation, even where this is merely recollection, and does not restrict it to mean the capacity for the new formation of concrete individual representations. MEUMANN is surely right when he holds¹³ that the narrower meaning of the word imagination first arose under the influence of the modern age’s more penetrating study of artistic production. Artistic imagination leads to works which to an observer can seem to be the fruit of a reflection that with full consciousness lays in the single features in such a way that they co-operate with other features to the formation of a whole. On the occasion of a feature which Solger had brought out in his analysis of the characters in “Wahlverwandtschaften”, GOETHE said: “Ich habe selber nicht daran gedacht, aber es liegt darin”. (*Gespräche mit Eckermann*. 21. Januar 1827.)

- [30] The artist thus comes involuntarily to say more than he is conscious of. That he analogizes, and thereby comes to connect elements that were otherwise separated, he does not notice until perhaps later, if testing reflection awakes. By a living-into — or, as the Germans say, a feeling-into (*Einfühlung*) — of the subject-matter, which involuntarily leads him to seek and to find expression for its value, he can awaken, if not an independent creation, then at any rate an independent capacity for a living-into like his own. In this latter living-into it is not only the subject-matter’s content, but also the form under which it comes forward in the work, that exercises a decisive effect — just as the artist’s living-into the subject-matter leads him from the first to a forming activity which might let the subject-matter come forward in its whole fullness. In the definition which JULIUS LANGE (*Om Kunstværdi*. 1876) gave of artistic value, as the value the subject-matter has for the artist and which it acquires through his work for the beholder, the form

¹³*Intelligenz und Wille*², p. 136.

of presentation did not come into its own (as I sought to show in a review of Lange's fine book in "Nær og Fjern" 1876); this Lange regarded as something purely external and mechanical. In the artistic genius, need (feeling of value) and capacity (forming power) are joined from the first — and joined in the most intimate way, since there is a need to use the capacity, and not merely to feel the value, and since the feeling of value presupposes a capacity for opening oneself to what can call forth that feeling. FRIEDRICH VISCHER has rightly maintained that the feeling-into must determine both the work's form and its content. Lange, by emphasizing the living-into of the subject-matter, pointed to what is involuntary in the imagination and in the genius; but to this belongs also the involuntarily forming activity. On Lange's definition it might look as if the work of art [31] were merely an external, inessential means by which others than the artist himself could experience the subject-matter's value. Need and capacity are not bound together in so outward a way, even though the carrying through of the forming activity can at many points make possible, indeed require, conscious deliberation and choice.

The concept "Einfühlung" was known neither to Lange nor to me when we conducted our discussion. It had been set up some years earlier by the German aesthete ROBERT VISCHER, but became more generally known only through VOLKELT's and FRIEDRICH VISCHER's writings.¹⁴

While for the primitive consciousness the image stood as the bearer of a magical power which made its possessor master over the subject-matter, in art it is the bearer of a value that was born in the artist's soul and through his work is born again in the beholder's soul. In both places there is an analogy which works in an involuntary manner, and which is seen as analogy only by testing reflection. That the involuntary mental life's manner of working has not been understood led to a failure to recognize the nature both of the genius and of primitive man. But the understanding that can be won of the involuntary mental life comes about itself by the help of analogy, since the understanding starts out from the differentiated life of consciousness, which can make the primitive syntheses clear to itself only by dissolving them.

¹⁴JOHANNES VOLKELT: *Der Symbolbegriff in der neueren Ästhetik*. 1876, p. 55. — FRIEDRICH VISCHER: *Das Symbol* (Festschrift til Zeller. 1887) p. 170–187. — ROUSSEAU knew feeling-into at first hand. His joy in nature springs from it. Cf. *Confessions* (éd. Bever) I, p. 263 f.: Je dispose en maître de la nature entière; mon cœur, errant d'objet en objet, s'unit, s'identifie à ceux qui le flattent. — But in Rousseau the need was greater than the capacity. A work of art he could not form (apart from the work of art his own *Confessions* are). But his heartfelt feeling gave the impulse to modern poetry.

[32] Answering to the image's magical significance there stands, in art, its symbolic significance; but in the history of religion magic and symbolism are broken wholly apart only in the most recent time.

5. The symbol rests on participation, in that a part stands as the expression of the whole and works as such. The small can become a symbol of the great when it presents in itself relations that have a likeness to relations within the great. For the great humour, everything single, however limited it may be, can become a symbol by being seen as part of a great whole and as presenting the same type as the whole. "The humblest thing defends its place in life's garland." But life's garland is never wholly bound, and the possibility is always present that new flowers are to be woven in. To this the great humour will not be blind; it looks humorously on its own garlands,¹⁵ and it agrees with GOETHE (*Gespräche mit Eckermann*. 5. Juli 1827) that in images used as expressions for life there always remains something incommensurable for the understanding. The image's value rests precisely, on one essential side, on its giving as a whole what for testing reflection stands as an unclosed connection of elements — perhaps of constantly changing elements. Seen from this side, the image gives less than the always advancing experience and reflection. As a whole-image it gives more, since as such it has elements not to be found in the subject-matter from which it is fetched. When God is called Father, the element of omnipotence comes in with it, which unfortunately is lacking in every father given in experience, who must often struggle hard both for himself and for his own. Or (to use FR. VISCHER's example) when the lion is a symbol of magnanimity, it has a property the real lion lacks. Pseudo-hallucinations are therefore often better suited [33] as a foundation for symbols than hallucinations proper, which as a rule take in all the elements of reality.

Religious symbolism is especially characterized by the fact that its foundation of feeling has formed itself under the struggle for what, for the human group in question, is the highest thing of value. The images in which expression is found for the experiences made in the struggle for existence are fetched from the great oppositions of nature and of life: light and darkness, life and death, joy and sorrow, power and powerlessness, good and evil; and, as the most elementary nature-religions give place to higher forms of religion, a need makes itself felt to see these oppositions represented and working

¹⁵*Den store Humor*. Chap. 6 (Forstaaelse og Humor).

through more or less individualized beings.

But like the concept of analogy in general, the concept of symbol too is first formed at the standpoint of testing reflection. The difference between part and whole, between the sign and the signified, does not make itself felt from the first. What for the artist and the artistic beholder holds only in moments of enthrallment is, for the positively religious consciousness, so long as it subsists in its full peculiarity, the constant state. Every distinction between part and whole, image and subject-matter, is heretical. Hegel wrongly held that in all symbolic apprehension a dissatisfaction must be felt on account of the ever-present disproportion between form and content. Since Hegel presupposed such a dissatisfaction already at primitive stages, and in general within the positively religious consciousness, it became comparatively easy for him to show that it was a liberation for the religious consciousness to pass over to forms that could subsist for testing reflection; and the difference between religion and philosophy became for him in the end only a difference of form, while the spiritual content was to be the same.¹⁶ Hegel [34] reads the cleavage to which reflection can lead into the religious consciousness itself, for which, on the contrary, the relation between part and whole, sign and subject-matter is a relation of identity. It is denied, e.g., from the side of positive religion, that the expression Father used of God is figurative. So far is it from being experiences of the significance of the human father-relation that are to lie at the base when God is called Father, that the case is precisely the reverse: it is God who is the ground and the model for every father-relation in the world (the Epistle to the Ephesians III, 15). By virtue of this identity it is possible that actions which for reflection stand as symbolic — such as prayer, sacrifice and sacrament — can be assumed to exercise an influence on the Godhead. This is consistent, for by virtue of the relation of identity there is a real connection between sign and object. Added to this, the positively religious symbolism is the result not only of the individual's but of a whole human group's experiences of life, and thereby comes forward for new generations as the given and self-evident expression of what can be experienced about human life and its conditions.

Concepts such as analogy and symbol are correlative concepts; they express a reciprocal relation. As we have already seen with the father-symbol, the relation therefore turns quite naturally round. From the world of experience in which human life is led,

¹⁶ *Den nyere Filosofis Historie*.³ III, p. 182–186.

symbols were fetched for a higher world; but in the end the very world from which the symbols were originally fetched becomes a symbol. In the mystics of the Middle Ages this turn comes forward in a characteristic way. For Hugo a St. Victore the whole bodily [35] world is a symbol of the spiritual world. And when Goethe uses medieval expressions to denote the soaring above everything finite, he declares

Alles Vergängliche ist nur ein Gleichniss.

Medieval thinking moved, as it has aptly been said, as often from symbol to symbol as from experience to symbol. There had once for all been awakened a fundamental mood which constantly sought new expressions, and in relation to which different experiences of life could mean the same. Here the emotional foundation is at work which is in general determinative for all “participation”, all “prelogical mentality”. Only critical reflection makes that demand on religious representations which Schleiermacher maintained, that each single symbol shall be formed by itself on the foundation of a particular lived experience. In the doctrine of faith, he says, the one proposition must not be derived from the other, but is to be set up immediately with reference to a single determinate lived experience, and only then afterwards compared with other, previously established propositions. (Der christliche Glaube § 17.) So long, on the other hand, as the once developed fundamental mood maintains itself, new symbols will spring again and again from it, the one developed in analogy with the other. Thus YRJO HIRN (*Det heliga Skrinet. Studier i den katolska Kyrkans Poesi och Konst. 1909*) has shown in a fine way how the heartfelt worship of the Madonna leads to a series of images which express the fundamental mood from different sides: the dwelling-place of the Host in a shrine on the altar, the place where the divine comes forward in bodily form, became a symbol for the Madonna, in whose womb God became man. But the Madonna is at the same time the morning star that heralds the sun’s coming, and she is the spring, where light and warmth relieve the darkness and cold of winter. The mighty expansion of feeling constantly puts out new shoots.

[36] The involuntary formation of images is the presupposition of all dogmatics. Not all symbols become dogmas. There is talk of dogmas only when certain symbols stand as norms which individuals ought to follow in their attempts to interpret their experiences of life. A choice is then made, and what is chosen stands fast in virtue of the religious

community's authority over its members. The dogma rests on a mixture of lived experience, reflection and authority. From what has been dogmatically fixed a struggle is then carried on against freely developed symbols, as well as against the attempts to apprehend the dogmas themselves as mere symbols. Dogmatics' struggle against symbolism is a struggle between "it is" and "it means".

A free formation of symbols within a religious frame is presented by Jesus' parables, in which through narratives or features seized out of life he makes intuitable the relations and the demands of the promised kingdom of God. An ethical-religious wisdom of life is here uttered in poetic form. The figurative is chosen with deliberate intent, and often (especially in the parables of the Gospel of John) passes over from symbolism to allegory.

The symbol becomes allegory the more expressly a distinction is drawn between the two members in the symbolizing. In allegory the relation between the two members is more external than in the symbol. Poetic symbolism springs, like religious symbolism, from the enthrallment a subject-matter has awakened; but it does not exclude, as religious formation of images in the classical ages of the religions does, that the enthrallment may alternate rhythmically with a clear consciousness of the formation of images that has taken place.

In intuitive natures, where intuition and imagination have the preponderance over reflection, there will be a tendency to live in symbols as long as possible. Still stronger will [37] this tendency be in ecclesiastical natures, for whom it is a need to hold fast to inherited symbols — not only on account of their intuitability, but because they are the inherited expression of the fathers' faith, and thus on account of a feeling of solidarity with the past. They hear a centuries-old tone in the church's hymns and find again the race's dearly bought experiences in the church's dogmas. Yet it must be noted that it need not be intellectual criticism alone that bursts the inherited forms; it may also be a new and strong movement of feeling that makes it impossible to find one's own experiences of life expressed in these forms.¹⁷

¹⁷Cf. my *Religionsfilosofi* §§ 95–96. — *Oplevelse og Tydning* p. 102–107.

Chapter II

Analogy and Logic

6. In the involuntary analogies, likenesses and differences are from the first so woven into one another that it can be difficult to find the boundary where the likeness ends and the difference begins. The definite attempt to find such a boundary, through a sharp determination of the different degrees of likeness (and of difference), marks the beginning of scientific thinking. Plato stands in the history of thought as the first who drew attention to there being many steps upward from the likenesses that come forward for immediate apprehension to pure identity, the “Idea of likeness”, which alone makes strict science possible. What Parmenides had proclaimed, under the name of “the One”, as the sole condition of true knowledge is determined by Plato more exactly as identity. — In my treatise on the Platonic dialogue *Parmenides* (Vid. Selsk. filosofiske Meddelelser Nr. 3) I express myself perhaps too strongly when I say (p. 28) that Plato “expressly declares” [38] unity and identity to be the same; more cautiously it is said on p. 30 that it is “let out” that unity and identity are the same. The fact is that Plato in the strange second part of the dialogue apparently denies this; but since he grounds the denial only on the point that *several* things (τὰ πολλά) can be One (i.e. can come under the same concept) without coinciding, and adduces no proof at all in the case of “the One”, which is just what is in question, I find here an involuntary concession. So much is certain, that Parmenides’ “the One” is obscure and mystical when it does not mean identity — that is, when it does not express at once the fundamental law of all thinking and the highest ideal of all thinking.

For Plato, now, the Idea of likeness was not merely fundamental law and ideal, but the expression of true being, and the degrees of likeness were so many lower and higher

steps of existence. All the same, in the setting up of this Idea a direction was given for the dissolution of the involuntary analogies. When that dissolution is carried out with a hard hand, absolute likeness and absolute difference come to stand in opposition to one another, identity in opposition to chaos, and it becomes cognition's task to find a road between these oppositions. It is characteristic of the sharp Greek thought that not only the Idea of absolute identity but also the Idea of absolute difference is brought forward, albeit with a polemical purpose (by Zeno). And towards the close of his career Plato saw that the principle of identity stood in the end as powerless in the face of the manifold differences that existence presents, and therefore refers to analogizing as a kind of "spurious thinking" (λογισμὸς νόθος. Tim. p. 52 B). Analogy stands here, then, as a last [39] resort, where no higher degrees of likeness can be exhibited. Whereas in mythological thinking it was everything, it now comes to denote the lowest stage of cognition. The programme of European thinking was herewith given. And the task then becomes to exhibit the different intermediate forms between chaotic difference and absolute identity, for it is in these intermediate forms that human thinking actually moves. The order of rank of these intermediate forms (of which I have described six in *Den menneskelige Tanke*) will depend on how far the subject-matters before us can be ordered in such a relation to one another that inference becomes possible. It will then show itself that the more inference becomes possible, the more we approach pure identity. It shows itself at the same time that identity no longer stands as the highest form of existence, but as the form of thought which makes grounded cognition possible without calling in the help of observation at every moment. The principle of identity dawns most clearly in such series as those where each member stands to its predecessor as that one does to its own predecessor. If, e.g., *A* is subject to *B* and *B* to *C*, or *A* is smaller than *B* and *B* smaller than *C*, then *B* can be eliminated and we get a new judgment. The presupposition is *B*'s identity with itself, whether it be set in relation to *A* or to *C*. But such a presupposition holds for every member in every series we set up. To that extent the principle of identity lies at the base of all series-formation; but cognition consists not merely in the formation of single concepts, but in the exhibiting of intelligible relations between concepts, in rational series-formations.

The expression "participation", so happily chosen to denote involuntary analogy, can well be retained after logic has begun to have a word to say concerning the presuppositions

of the course of thought. It denotes a partial likeness which in the involuntary analogizing was made into an absolute likeness, in spite of the difficulties there are in grounding the [40] right to assume, from a likeness at a single point, an agreement in general. It is precisely these difficulties in “participation” that set scientific thinking its tasks.

If one seeks a presupposition which for all that lies at the base both of prescientific thinking’s involuntary analogizing and of the scientific search after grounded transitions, and which lies at the base of all series-formations from the chaotic difference-series to the absolute identity-series, then such a presupposition lies in the law of synthesis, which holds both in empirical psychology and in rational epistemology. It lies in the concept of a series — whether the members are held together by mere memory or by grounded understanding of the transitions — that different members are gathered together with one another, either as different or as alike. If thought constantly lost the preceding members when it went over to the following ones, no series-formation would take place. The more it becomes possible to ground the transition from member to member, the smaller the role that mere memory — which is sole ruler in the chaotic difference-series — needs to play; the transitions then take place in virtue of a law.

The fundamental law of synthesis rules at all stages of the connection of representations, and thereby it becomes possible to compare these different stages and to exhibit the relation between them. If the different sciences, which find themselves at different stages between chaos and identity, did not have such a common fundamental form, we could not set up the very concept of science as common to them all. We can now elucidate not only one science through analogy with other sciences, and one part of a science [41] through analogy with other parts, but also prescientific thinking through analogy with the scientific; and we can within psychology elucidate the elementary mental functions by analogy with the more developed ones. Such analogies are not mere fictions, as VAIHINGER held in his *Philosophie des Als Ob*.¹ In virtue of the continuity which through the fundamental form of synthesis makes itself felt in all life of the spirit, and especially within the intellectual life, analogies such as those mentioned become possible; naturally their logical value is highly various. It is, e.g., no fiction when a tangent is called a secant whose two points of intersection with a circle approach one another in the highest degree;

¹An interesting presentation and criticism of *Philosophie des Als Ob* has been given by JØRGEN FR. JØRGENSEN in the journal “Vor Tid”. 1914–15. p. 227–249.

for the synthesis can bring its elements together in a great many degrees. According to a modern tendency in geometry the tangent even always touches the circle in more than one point, so that the difference between secant and tangent falls away in the end.

7. Even between the two outermost limits of our cognition, the chaotic difference-series and the absolute identity-series, analogy is present. The only operation of thought they make possible is — apart from useless substitutions — counting. One can count how many absolute differences, or how many absolute identities, one knows. As we shall later see, arithmetic takes hold precisely where pure logic ends. Another analogy between them lies in this, that neither of the two series can be immediately given. Only through a strictly critical testing, which could not be shown to be exhaustive, could it be shown [42] that there were no points of likeness at all, or no differences at all, within a series of subject-matters. The two series mark limits of our cognition which cannot be proved to be present in any single case whatever; only through arbitrary assertions could their validity be maintained. But these series serve indirectly to characterize human cognition, by exhibiting what would make it impossible from the first, or would be a conclusion admitting of no possible continuation.

One might say that epistemology has only the task of discussing the principles that make cognition possible, and that it must therefore first of all set up the principle which lies at the base of all formation of concepts and judgments and inferences whatever, namely the principle of identity. Cognition consists of judgments, one might say, and every judgment asserts a relation of identity (partial or absolute). The concept of identity would therefore seem to have to open the series of the categories. To this it must be answered that epistemology ought to investigate all possibilities. Every actual cognition is always only an approximation to identity, and all degrees of approximation must be investigated. Epistemology's proper subject-matter is the degrees of clash between identity and difference. Identity itself has only this significance, that it sets the advancing reduction of the differences as the great task. Identity is, after all, itself a relation. To strive after finding identity is therefore to seek to put one relation in the place of another. The concept of identity thus presupposes the concept of relation and the concept correlative with it, synthesis, and these become the first categories. Besides, identity can be defined only as a limiting case of likeness or of continuity, and these two concepts therefore go —

when we proceed analytically² — before the concept of identity.

When it is said that cognition consists of judgments, it must not be forgotten that the judgment's subject is always taken out of a whole connection, and that this taking-out becomes possible only through the difference there is between it and other subject-matters that are members of the same connection. Only by such a distinguishing does a subject-matter come to exist for us with such independence that closer determinations of it can be sought and found. As it has aptly been said, every concept has fringes or rents which show that it is properly a fragment. Only when this separating-out has been carried out can a judgment come into being. If every subject-matter drowned in, or were wholly fused with, the medium in which it exists from the first, no judgment would ever be passed. A relation of difference is both the logical and the psychological presupposition of a judgment's coming into being. [43]

Naturally the principle of identity is presupposed in all logical thinking, also in the setting up of a doctrine of categories, and especially in the setting up of the categories synthesis and relation. Epistemology must of course build on the principles all science builds on; and since it is precisely these principles that epistemology wishes to bring forward and elucidate, it can to that extent be said to build on itself, or to move in a circle. Occasion has sometimes been taken from this to demonstrate the impossibility of an epistemology. So Schelling, Lotze and, in the most recent time, LEONARD NELSON (partly in his work *Über die sogenannte Erkenntnistheorie* 1908, whose first section is entitled *Die Unmöglichkeit der Erkenntnistheorie*, partly in a lecture at the congress of philosophers at Bologna in 1911). Nelson's train of thought is this: the criterion of cognition's truth must either be cognition or not be cognition. If it is cognition, then this cognition must itself require a criterion. And if it is not cognition, it must all the same be something [44] that can be an object for cognition, and then we require a criterion for that cognition's being true. — This train of thought we know already from antiquity. It was brought forward on the sceptics' side against the Academics' attempt to set up a criterion of reality, when they asked by what criterion we can decide that the criterion of reality so set up is the right one!³ At the base of this scepticism, as of the modern objections to the possibility of epistemology, lies a confusion of the concepts of circularity and

²Cf. *Den menneskelige Tanke*, p. 177 f.

³See more on this in *Relation som Kategori*.

incompleteness. Epistemology's task must always be taken up anew. It must always be tested anew whether the principles which epistemology sets up as conditioning the validity of cognition coincide with the principles that lie at the base of the very attempt to set up the epistemology. If there is a difference here, the investigation must begin over again. A conclusion of these investigations is not to be thought. It may very well be that a new investigation of human thought-work can lead to the necessity of a new epistemology. Before Carneades too, and in modern times before Hume and Kant, men had an epistemology; but it showed itself that it had not been carried through, and that its fundamental significance had not been demonstrated.

Epistemology comes into being through the investigation of thought-work itself, that is to say, of the alteration which the application of certain principles brings about in respect of clarity and connection in our representations. If we find advancing clarity and connection, we say that epistemologically a step forward has been made, even if we perhaps come to strike out certain assumptions on which we formerly laid weight. The [45] more numerous and the deeper the differences between the subject-matters that can be brought out, and the more we nevertheless succeed in finding grounded connection between them by the application of the principle of identity, the more we become convinced of that principle's validity. As we have seen, the principle of identity can be said to dawn already in the prescientific analogies, as well as in those series-formations which do not yet form a foundation for inferences. When, as in a regularly varying difference-series (e.g. the colour-series), each member lies fast, so that no interchange can take place, the principle of identity has come more into force than where the order is either quite arbitrary (as in the chaotic difference-series), or where the order is due to a wholly external point of view, e.g. a scheme for the memory (as in the indeterminately varying difference-series). And the history of the sciences shows us many stages in this respect before full clarity and certainty are attained.

8. In the analogies that play so great a role in prescientific thought-life there already stirs the energy which can later lead to scientific thinking. They ground the expectation with which men come to meet the subject-matters that present themselves to them, and which makes them not stand passive in the face of what observation offers. With a sanguine confidence in all emerging representations, wherever they may come from, the life of thought begins its course of development; indeed, even before free representations

appear, an analogy to this confidence declares itself in the involuntary propensity to immediate motor reaction, to which reflexes and instinctive actions testify. Life begins with a surplus of energy and has a tendency to spread and unfold itself. Before memory in the narrower sense of the word can form itself (with a clear consciousness that what is represented belongs to the past), expectation can be at work; and representations [46] which are not bound up with any expectation, but whose content stands definitely as past, are held fast only when a higher development has been reached. Hume rightly laid weight on the need for expansion, for paying out the line, for continuing a train of thought one has once entered upon. But when he would derive from this tendency the belief in rational and causal principles, he overlooked that these principles come forward most clearly when expectations are disappointed or rejected, and when we are then compelled to seek an order and connection of another kind than the one our involuntary analogies and expectations had announced. Rationality and causality are norms which limit and dam in our involuntary course of thought, which in itself has a tendency to pursue certain ways of looking at things beyond all limits. The hypotheses which science on its way has shaped, but has then let fall again, testify that it is for a great part through disappointments that we work our way forward to truth. If disappointments had not come, we should at many points not have got beyond the prescientific stage of development; but the disappointments would not have come if the surplus energy had not fostered expectations which were involuntarily tested by our acting on them. Children's need to grasp and work upon all things leads them to make important experiences. They experiment involuntarily. The need for movement enters by degrees into the service of curiosity. In his autobiography the German surgeon Schleich relates that at five years old he had already been given a watch, but smashed it by knocking it with a stone; and when he was then scolded for it, he said: "Jungens müssen doch wissen, wa da'inn is!" This need to know what is "inside" stirs from the childish experimenting all the way up [47] to the highest speculations, without there always stirring any intimation of how fine and how complete is that connection of things into which we seek to penetrate. That need is by degrees dammed in, as it is experienced that determinate laws rule for things, and that it is through the understanding of these that we may perhaps penetrate into what we call nature's interior.

In a spirited work (*De la contingence des lois de la nature*. 1875) ÉMILE BOUTROUX

developed the conception that the laws of nature signify essentially forms or channels for the unfolding of free forces — the river-bed for a stream that works its way forward. A step further went HENRI BERGSON, in demanding of philosophy that it should penetrate behind the mechanism of the laws of nature and make itself one with the force, the unfolding of life, which is constantly hampered and limited by the order of nature; and he found a foundation for this in an immediate intuition of our own inner unfolding of force (*élan vital*), in analogy with which we must then apprehend all other forms of existence than our own (Introduction à la Métaphysique. 1903). — Such attempts of thought are in reality due to a translation into metaphysical form of what psychology and epistemology teach. They are good examples of what we shall in a later connection go into somewhat more closely, that all metaphysics (cosmology) builds on analogy — here on analogy with the way in which we see, in the history of science, the human impulse to expansion and human impatience clash with the fixed order of things which science works to find. Of special interest here is a chapter in ERNST MACH's work *Erkenntnis und Irrtum* (1905). The chapter is headed "Sinn und Wert der Naturgesetze" and seeks to show that the laws of nature come into being for us as limitations which, under the guidance of experience, [48] we set to our expectations, and that through such advancing limitation there takes place an ever closer adaptation of thought to facts. Even daily experience shows us a constant damming in of the involuntary life of thought.

9. Just as the cognition of laws is a limitation of the freedom to form connections of representations, so the formation of concepts is a limitation of the freedom to shape the single representations. We form concepts in order to be sure that a representation which emerges anew in our train of thought has the same content as when it appeared formerly; only thereby can it enter as a fixed member in a series of thoughts and stand in clear and determinate relations to the other members of the series. If it is merely a matter of holding fast what is peculiar to a subject-matter in a quite determinate situation, this is attained by what one might call a concrete individual concept. Thus "Napoleon at Waterloo" is a concrete individual concept, when we have noted Napoleon's whole conduct on that occasion. If the concept is to hold for all sides of a subject-matter's essence and for its appearance in all different situations — e.g. for Napoleon according to his whole character and course of life — we get a typical individual concept. Each single situation or each single action then serves only as an example of this concept.

And in a similar way it is with general concepts, which are to hold for a whole group of subject-matters. The “common” properties will here, just as little as with the typical individual concepts, be sufficient to be the content of a concept by itself. The “common” cannot be thought by itself alone. More or less distinctly, on the other hand, a single subject-matter stands as an example of the whole series; or one makes use of a scheme [49] or a word which, if it be needed, can be replaced by an example.

Between the different examples themselves there will not be complete identity in all respects. It is some point of likeness (perhaps of identity) that leads one to set subject-matters together as examples of the same concept. An Indian who saw a book open for the first time called it an oyster. That was already an attempt at a formation of a concept. When men early spoke of feet both in animals and in men, and later extended the concept to tables, chairs, mountains and so on, that too was an attempt at the formation of a concept. But not even in scientific formation of concepts do the single examples completely cover one another. It is by the help of analogy that one goes from example to example within a concept’s extension. Under the concept of the conic section belong circle, ellipse, parabola, hyperbola, and the relation between these subject-matters is not identity but analogy. Only from a determinate point of view, which the definition of the conic section states in words, can they be regarded as identical, in so far as any one of them can in principle be used as an example just as well as the others. The very definition of the conic section (as a figure that arises by a plane section through a circular cone) requires a construction which leads to one or another of the examples adduced. It is the relation between properties or parts in a series of subject-matters that makes these subject-matters examples of the same concept. With the concept of colour there is provisionally only qualitative likeness between the single examples, and a concept proper is got only by keeping (as with the conic sections) to the way in which they arise. Each single colour-quality stands to a certain degree of refraction of light as the other colours stand to certain other degrees of refraction. For brevity’s sake (when one has [50] not time to bring forward examples) words or signs (schemes) can express the concept. Thus letters in arithmetic are a scheme for the concept of magnitude, whose examples are had in the members of the number-series. “Mathematicians”, says PASCAL (*De l’esprit géométrique*), “and all who proceed methodically, give things names only in order to shorten the course of thought”. The word or the scheme is to give directions for the

construction of an example. Pascal requires that for safety's sake one should try to put what is defined in the place of the definition. But it must be remarked on this that, since "what is defined" comes forward in different shapes, this substitution can never become absolute. No example contains merely the features the definition states. Each single example requires other examples, and it is the analogy between them that is the concept's proper content. The concept's significance consists in its being able to set a series of thoughts going. GOBLOT therefore says that a concept is nothing but a possibility of a series of judgments (*une virtualité, une possibilité indéfinie de jugements*).⁴ He therefore begins his exposition of logic with the judgments and goes from there over to concepts and inferences. Earlier still, SIGWART had maintained the priority of judgments in relation to the concept. But the matter is surely more involved than the two excellent logicians present it. There is a constant reciprocal action between concept and judgment. The formation of concepts itself takes place through judgments, but the judgments are clear only when the concepts connected in them are each determinate. The history of thought shows us how the starting-point is taken now in the single concepts, now in [51] the judgment. In epistemology this comes forward in the relation between fundamental concepts (categories) and fundamental propositions (principles).⁵

There is something in our thinking (and in our consciousness in general) which never comes forward for immediate observation, but which must be presupposed or, if one will, constructed, because what immediate observation exhibits is otherwise unintelligible. Thus synthesis, the fundamental form of conscious life and the first fundamental concept of all cognition, is not itself an observable element. Nor can our willing as such be immediately observed.⁶ And so it goes with all inner and outer activity: we observe its results and infer to the energy to which the work is due. The activity of thought itself is not observed. Every single thought is a member in a course of thought, and we infer to thought's energy from those alterations in the course of thought which are independent of momentary observations (except in so far as the alteration consists in such observations being brought into determinate connection with the previously given content of consciousness). Everywhere we observe only succession, not the activity which stretches through all the members of the succession and which we often have

⁴*Traité de Logique*. 1918. p. 87.

⁵Cf. *Den menneskelige Tanke*. § 51; 57; 106.

⁶*Psykologi*⁶ p. 177 f. (Forkortet Udgave p. 33 f.; 84 f.)

laboriously to trace out. Here Hume's criticism still holds good, in the domain of the human sciences as in that of natural science. His violent isolation of the single members in the course of thought led indirectly to the right apprehension of the concept of activity in all domains. But it is Leibniz who has directly indicated the right application of concepts such as force, energy and working, in maintaining that before there can be talk of force a lawlike connection between changing states must have been exhibited.⁷

Thinking proper is distinguished from the ordinary course of representations only [52] by being directed towards the conditions for solving a task, towards means and ways of reaching a goal. It consists in an analysis of the connection within which the task or the goal is assumed to have its place, and then seeks to find a law among the elements to which the analysis leads. A simple example is presented by orientation when one has lost one's way. The different directions are tried and compared. If, on the other hand, one wanders without a determinate goal, one strikes into the first road that offers. Already HOBBS — who so often has to be the scapegoat in those criticisms of "association psychology" of which people seem unable to tire — saw this, when he declares that it is through constantly holding fast to a goal (*frequens ad finem respectio*. De Corpore XXV, 9) that we learn to find order and connection.

10. The concepts of analogy and of inference from analogy were first brought out with clarity by Aristotle. Whereas for Plato, even after he had had his last single combat with the Eleatics, it stood as the great goal to lead all cognition back to one Idea, Aristotle is convinced that we end in a plurality of concepts (categories), between which not identity but only analogy holds. Between the different domains in which thinking works, analogies too can perhaps be exhibited; thus Aristotle holds that the relation between matter (possibility) and form (actuality) recurs everywhere — but this relation itself takes on a different character in the different domains (that is to say, within the different sciences).

As regards the conducting of proof, Aristotle distinguishes between deduction and induction. By deductive inference he understands the syllogistic bringing of something under a genus-concept by the help of a species-concept as intermediate member. His [53] doctrine of inference is thus built on a presupposed classification. By induction he understands a complete enumeration of all the subject-matters that come under a concept

⁷Den nyere Filosofis Historie.³ II p. 136 f.

(ἡ ἐπαγωγή διὰ πάντων. ed. Bekker p. 68 b). But there is also, he adds (p. 69 a), an inference from a single example, or rather from example to example. This inference thus goes neither, like deduction, from a whole to a part, nor, like induction, from all the parts to a whole, but it goes from one part to another part (ὡς μέρος πρὸς μέρος). It is distinguished from induction in that not all the parts, but only a single part, lies before us. Aristotle's designation for inference from analogy is Paradigm (παράδειγμα), which means example, but can also mean model. The single subject-matter can by inference from analogy be said to stand as a model, a type, in whose light a whole circle of subject-matters is seen, in that they present like relations between properties or parts among themselves. — The word ἀναλογία itself Aristotle uses mostly in the sense of mathematical proposition, identity of relation (ισότης λόγων), while for qualitative likeness of relation he frequently uses the indeterminate expression τὸ ἀνάλογον.

In more recent times it has been seen more and more clearly that there is no difference in principle between induction and analogy, since on the ground of experience one can never be sure of having all possible subject-matters of a certain kind given.⁸ Incomplete induction is then distinguished from inference from analogy only by having more examples to start out from. Naturally it is of importance to have as many examples [54] as possible to start out from. But for modern methodology the main thing is whether, by an analysis of the subject-matters before us (one or several), a proposition can be found which can provisionally be set up so as then to be tested by seeing whether the inferences drawn from it find confirmation in new experiences. This method, which was first applied on a great scale by Galilei and Newton, is characteristic of modern natural science. Analogy and induction give, as KANT expresses it in his *Logic* (ed. 1801, p. 210), merely “logische Präsumtionen”. Strangely enough, he does not add that it may all the same often be of importance to test these surmises.

KANT himself saw clearly that there is a kind of inferring which is neither deduction, induction nor inference from analogy. He refers here to the principle by which mathematical thinking differs from formal logical thinking. In virtue of this principle an identically varying difference-series can be constructed, that is, a series within which the difference between the members following one another is always the same — where, accordingly,

⁸Except of course in such cases as those where it lies in the very definition of the concept that there must be a certain number of subject-matters. Cf. *Den menneskelige Tanke*. p. 172 f.

there is identity of difference. The simplest example of this is the number-series, which is built up by what Kant calls “Synthesis des Gleichartigen”. Besides in the number-series, the same principle appears — surely under the influence of the number-series as a model, and so in analogy with it — in the series of times, degrees and places, when (as with number) all special qualitative properties are purged away from these concepts. Judgments built on such series-formations (which may be called identical quality-series) Kant calls synthetic judgments *a priori*. In analogy with these series built on identity of difference, Kant has now constructed a concept of “possible experience” or “nature” as a scheme within which every subject-matter to which we are to be able to ascribe reality must find its place. We shall at a later point have occasion to discuss the significance of this [55] construction.

While we are on the prehistory of the concept of analogy, it must be pointed out that in philosophical attempts to reach a world-view, a concluding cognition, the dispute in modern times stands between those who would reach a conclusion through inferences, that is, by the help of the principle of identity, and those who maintain that analogy is the last means of thought when it is a question of conclusion. It is the opposition between Plato and Aristotle repeating itself in new forms. On Plato’s side stand Spinoza and Hegel, on Aristotle’s side Leibniz, Schopenhauer, Lotze and, in the most recent time, Bergson. From SØREN KIERKEGAARD’S *Papirer*⁹ one sees that he was aware of the opposition between identity and analogy at the decisive points, in good agreement with the critical tendency of his epistemology. He had become clear that a systematic conclusion was impossible by the help of the principle of identity; and since induction and analogy permit no strict inferring, he came to the result that one always concludes (as one begins) with a leap — and it was surely a satisfaction to him to find an application here for one of his favourite concepts. But the problem is still more involved than he saw, and it is precisely the leap that poses it anew: the leap takes place in a curve, and so brings about a new continuity, though on another plane.

11. When a manifold of subject-matters lies before thinking, its first step will naturally be to order them in series whose members are as closely connected as possible.¹⁰ The most [56] unfortunate relation would be that between all the subject-matters there subsist equally

⁹*Papirer*. Ed. P. A. Heiberg and V. Kuhr. V. p. 29.

¹⁰In what follows I make use of the attempt to set up and order the most important forms of series-formation which I have made in *Den menneskelige Tanke*. p. 162–173.

great differences, so that they can be put in any order whatever. As already shown above, it would be wrong to believe that such chaotic difference-series were peculiar to the primitive consciousness; this is characterized precisely by an involuntary interpretation, which lies at the base of the reflection it is able to institute. Only for a critical investigation, which is not content with any interpretation that offers, can it look as if the given were a chaos. As we have seen, the reality of a chaotic difference-series could not be proved. But the provisional assumption of a chaos can contain an invitation to closer investigation, and here analogy offers a starting-point. What would be common to the members of such a series would be precisely the circumstance that they are elements in a chaos; that holds for *A*, for *B*, for *C*, the whole series through. Herein lies the possibility of an inference in the Aristotelian second figure, one of the ways in which an analogy can be formulated. No positive inference can be derived from this figure, but there may be occasion to investigate whether after all there might not be other points of likeness between *A*, *B* and *C* than the one meagre point of belonging to a chaos. The second figure has in general the significance of setting a task. Already in this simple example, then, we see analogy's significance for thinking, as spurring, provoking and awakening. Analogy asks; but whether it answers is quite another question. And its very spurring effect presupposes that the participation has been dissolved, for by participation likeness at a single point is precisely made into likeness as regards the whole.

[57] We do not get much further when the members lie fast and cannot without more ado change places, as when the order of succession is determined purely chronologically. It would have to be that the question were expressly raised why the subject-matters were experienced or remembered in just this determinate order. This too presupposes a wakeful reflection. A child who is taught the Mosaic story of creation might, e.g., ask why plants were created before animals, or how light could be created before sun and moon.

If we have a regularly varying difference-series, such as, e.g., the colour-series, then not only do the members lie fast, but a certain continuity also declares itself in the order of succession, in that the qualitative likeness to the neighbouring colour is always greater than that to a colour lying further off. And there can be analogous relations within the series, in that, e.g., violet may be said to stand to blue as orange to red. But above all the question can come up how this determinate continuous order is to be explained. Goethe

believed he could find an explanation by keeping to the qualitative analogies. But in spite of his protest, science has gone beyond this standpoint of quality by exhibiting an analogy between the degrees of refraction of light and the members of the quality-series. Instead of the analogy within the series, we get analogy between two series, of which the one is quantitative, the other qualitative. It will appear in what follows that an inference from analogy in this form is of decisive significance for the modern age's apprehension of existence.

If a series is symmetrical, that is, can be read in the reverse direction without the single members being interchanged or any member eliminated, there is a new possibility of analogy within the series. If (to use de Morgan's example) *A* is in conflict with *B*, and *B* with *C*, there is analogy between *A* and *C* with respect to their relation to *B*. The inference from analogy can be formulated in the second figure: *A* is an enemy of *B*, *C* is an enemy of *B*. The third figure can also be used: *B* is an enemy of *A* and is an enemy of *C*. But the analogy leads here to asking how the enmity between *A* and *C* more precisely stands. It may be that the common relation makes *A* and *C* friends. But it may also be that the common enmity is not enough to overcome their mutual indifference, or the enmity which for other reasons subsists between them. [58]

With a progressive difference-series (that is, one in which each member stands to its predecessor as its successor stands to it) and with partial identity-series (that is, such as those in which each member is predicate to the preceding and subject to the following, or consequent of the preceding and ground of the following), not only are inferences from analogy like those already brought forward possible, but inferences can also be drawn by the simple elimination of intermediate members. A progressive difference-series is formed, e.g., by the numbers, the times, the degrees and the places, which we came to discuss above from another point of view. A partial identity-series is formed when predicates themselves become subjects for new predicates, and the consequents grounds for new consequents. The two kinds of series are characteristic of the formal sciences (logic and mathematics). But, as we have already seen, such series also acquire significance for purely qualitative difference-series. If it can be shown that a progressive difference-series or a partial identity-series within the circle of our subject-matters runs point for point parallel with a fixedly ordered quality-series, so that to a determinate member in the one series there answers a determinate member in the other, then from a

[59] transition within the one series one can infer to a transition within the other. All exact empirical science rests on such a setting together of quantitative and qualitative series. Where such a setting together is possible, one can regard the quality-series as if they were purely logical or mathematical series. One can, e.g., regard the colour-series as if the difference between the colours consisted merely in the difference between the degrees of refraction of light; and one can regard unavoidable qualitative successions as if they were merely logical transitions from ground to consequent (in a partial identity-series), or mathematical transitions from one degree of refraction to another. But it is and remains analogies that are built on here, even though there has here been made the great advance, in relation to the inferences from analogy discussed above, that not merely is occasion given for questions, but determinate expectations and surmises are grounded — hypotheses, that is, are formed, which can be sought confirmed either by new members being looked for in the qualitative series, or by the quantitative series being shaped with greater exactness than before, perhaps even in such a way that a new quantitative series is constructed. Whereas certain irregularities in the orbit of Uranus could be explained from Newton's theory of gravitation, the irregularities in the orbit of Mercury cannot be explained from that theory, and EINSTEIN has then attempted an explanation from the electromagnetic theory, from which the theory of gravitation itself would then be derivable.¹¹

Whether the two series will continue to run parallel cannot be known with full certainty. However long a continuation of the quality-series may be, it does not vouch for it that new subject-matters may not arise which do not order themselves according to the [60] rule stated in the quantitative series. Already LEIBNIZ saw that a complete verification is an impossibility. "A hypothesis' success", he says,¹² "does not prove its truth". To the complete confirmation of a hypothesis there would belong what Leibniz calls a complete reversal (*un parfait retour*) of the train of thought, that is, a demonstration that from the quality-series one came back with necessity to this determinate quantity-series and to no other. One would have to be able to reverse the inference and make its earlier premisses the conclusion. An objectively correct inference can, after all, be drawn from wrong premisses, and science's first (or last) presuppositions are therefore difficult to establish.

¹¹EINSTEIN: *Über die spezielle und die allgemeine Relativitätstheorie*⁶, p. 70. — EDDINGTON: *Space, Time and Gravitation*. 1920. p. 123–125.

¹²*Nouveaux Essais*. IV, 17, 5. (Op. phil. ed. Erdmann. p. 397.)

A relation of identity is in any case not present here, though a dogmatic tendency leads again and again to assuming one, in order to be able to rest in a First (or a Last). Quality simply cannot be derived from quantity, even though the study of the quality-series can lead to the setting up of proportionally varying quantity-series. Quantity is itself a quality which (as with number, time, degree, place) lends itself to the setting up of progressive difference-series. Science aims everywhere, where it can be done, at forming such series, in order then by their help to find lawfulness within other series whose qualitative character cannot be reduced in that way. We shall later come to discuss some particular examples of the great analogy on which all modern exact empirical science rests, often without being conscious of it itself. We shall also come to consider more closely Kant's great merit of clarity on this point, as well as the misgiving that still remains in the face of his theory. Besides Kant, it is especially the famous physicist [61] Clerk Maxwell (in philosophical respects a disciple of Kant at second hand) who has maintained the significance of the concept of analogy in this connection. To him Ernst Mach has on essential points attached himself. —

In conclusion it is to be remarked that even where absolute identity-series can be formed, we meet analogy as a presupposition. It was the Eleatic ideal to reduce everything to identity. If that could be reached, all existence's fundamental subject-matters could be presented in the series $A = B = C = D$ and so on. But (as I have shown in detail in *Den menneskelige Tanke* p. 177–201) every absolute identity between real subject-matters presupposes a determinate point of view in relation to which they can be regarded as identical, and so substituted for one another. From the point of view which the law for a group of subject-matters states, these subject-matters can be regarded as identical, though they may have many other properties than those the law concerns. It is the same relation as that between examples of the same concept.

Chapter III

Analogy between Functions of Cognition

12. From a psychological point of view it becomes more and more impossible to assume sharp oppositions between the different stages of mental life. There shows itself a constant continuity between the different stages, and thereby an analogy between the different stages, and thereby again between the different mental functions, becomes possible — even if we cannot, as there was occasion to insist above, without more ado identify the functions of the higher stage of development with those of the lower. There is here a connection which defies the distinctions men have wished to maintain. If, e.g., the so often drawn distinction between matter and form in mental life could be carried [62] through, there would be a limit both to analogy and to continuity. But sensation itself already defies this distinction. That is seen already when we compare sense-sensations with the corresponding physiological processes (so far as we know them): what is physiologically a composite process comes forward in the sense-sensation gathered together into a relatively simple element. We trace here already the law of synthesis. But this element itself is determined from the very outset partly by its relation to immediately preceding elements, partly by its relation to elements appearing at the same time. And very quickly reproduced elements are woven together with it. An absolutely simple and independent sensation, which should be an absolutely passive (i.e. externally determined) phenomenon, cannot be exhibited. Only approximately can such a complete isolation be

exhibited. SALOMON MAIMON¹ already objected against Kant's doctrine of sensations as mere "matter" that an absolutely passive sensation was to be compared with $\sqrt{2}$, and so could be exhibited only approximately. We thus find already in the simplest sensation an analogy both to what at higher stages comes forward clearly as synthesis, and to that distinguishing, that mutual determination of differences, which is peculiar to reflection.

A transition between sensation and reflection is formed by memory- and imagination-intuition. In intuiting, not only relations of difference but also relations of likeness make themselves felt; they condition the fusion or the organization which every intuiting [63] bears the stamp of.² Memory and imagination present, like sensation, an analogy to the way in which reflection strives to bring connection between the subject-matters accessible to it. The involuntary expectations which are the first form of memory and imagination form an analogy to the hypotheses which reflection sets up by the way of analysis. — An intermediate form between intuiting and reflection comes forward in the involuntary alteration of an intuitional image, by which it can become articulated, can change direction and centre of gravity, without reflection needing to play any part. There will, especially in some natures (those called above the intuitive), be a tendency to hold fast to intuiting as long as possible and to keep reflection's analysis away. It is an example of the frequent confusion of identity with analogy when many psychologists read judgments into the involuntary functions of sensation and intuition. The judgment becomes necessary only when immediate intuiting cannot, not even through articulations and displacements, hold all the subject-matters that draw attention to themselves. And the judgment will always be an imperfect surrogate for the intuition, since reflection can hardly separate out all the elements contained in the intuition and bring them into the new form of connection which the judgment is. The single judgment will therefore, for its supplementing, require new judgments, so as thereby to get more decimals for the determination of the irrational relation between intuiting and reflection. — One and the same type governs, psychologically regarded, the development of the life of cognition through its different stages. More fully than there was room for here, I have sought to demonstrate this in *Den menneskelige Tanke*. p. 34–54.

It may have its importance to see that continuity, and an analogy conditioned by

¹Cf. *Den nyere Filosofis Historie*.³ III, p. 122.

²The difference between fusion and organization as two kinds of mental whole-formation I had occasion to bring out in *Den store Humor*. p. 15–34.

it, make themselves felt as regards other sides of mental life too. Answering to the opposition between sensation on the one side and memory, imagination and reflection on [64] the other, there comes forward in the domain of the life of feeling the opposition between elementary and ideal feelings. Elementary feelings hang together with vital sensations answering to alterations in the circulation of the blood, respiration and digestion, or with sensations answering to influences from without. Representations play no part, neither recollections, imaginings nor reflections. Even with such feelings differences of quality make themselves felt — e.g. between a feeling of strength and a feeling of lassitude, between a feeling of ease and oppression, between a feeling of freedom and a feeling of anxiety. With ideal feelings, representations (recollections, imaginings, reflections) play a greater or smaller part. Thereby the states get a more determinate character, a more or less pronounced direction. The indeterminate feeling of ease and freedom becomes joy in that which rightly or wrongly stands as its cause; the indeterminate feeling of anxiety becomes fear of something that stands as threatening, and so on. But though we have here different stages in the development of the life of feeling before us, there is a constant transition between them. Ideal feelings are never independent of our organic states or of the sense-sensations that emerge at the moment, but may precisely be predisposed by these. Common to everything we in psychology call feeling is the difference between pleasure and displeasure; it holds for the ideal as for the elementary feelings, and these are only symptoms that feeling is present at all. No psychologist takes it into his head to derive the special qualities that come forward in elementary and ideal feelings from the abstract concepts of pleasure and displeasure. But it has an interest in this connection to note the analogies between elementary and ideal feelings, as well as the way in which [65] the transition from the former to the latter takes place. To apprehend as original the pronounced qualities which only the ideal feelings present would conflict with what genetic psychology teaches, especially the investigation of the life of feeling in the animal and in the infant.

In his interesting book *Følelseslivets Grundelementer* (1920) CARL JØRGENSEN has given a series of good descriptions of different feelings. One notices the good observer all through. But he lays no weight at all on the more indeterminate (and yet qualitative) feelings which I call elementary, and he denies that representations play any part in the development of the life of feeling towards more pronounced forms. And in his criticism

of my conception he overlooks (1) that I already assume, in the case of the elementary feelings, certain differences of quality (which therefore are not due to the influence of representations); (2) that in my terminology I understand by joy *joy in* something (a “joy” that was not joy in something I should reckon under the feeling of power, of ease or of freedom); (3) that I do not assume that it is due to representations that feelings acquire quality at all, but only that under the influence of recollections, imaginings and reflections there arise what I have expressly designated as “*new* qualities of feeling”. (See the fifth edition of my *Psykologi*, used by the author, p. 304; 305; 340.) I cannot recognize that the author has taken pains to enter into my conception; had he done so, he could not have christened it “the pleasure- and displeasure-theory” and given it the appearance that I assume feelings of pleasure and displeasure without any more special qualities whatever. —

In the domain of the life of will (when we take the word will in its widest sense) we also find a series of stages: need, impulse, deliberation, intention, decision. The [66] analogy between them rests on the active and concentrated character of the mental states determined by them. (On this I refer to my treatise *Begrebet Villie* in the third series of my *Mindre Arbejder*.) —

In all the domains of mental life, qualitative differences make themselves felt between the different stages. The one stage cannot be reduced to the other; but there are analogies between them which testify to the dominion of one and the same type.

13. It is one thing that within psychology it can be possible to find continuity between the different stages of development in spite of their qualitative difference, and that analogies can therefore be exhibited between more elementary and more elaborated mental functions; it is another whether on this point there is not a difference in principle between the psychological and the epistemological consideration of thinking and cognition. In the opinion of some, quite other principles must be applied when one asks about our cognition’s validity than when one asks about its actual constitution and development. I shall not here come back to the general question of the relation between psychology and epistemology, which I have discussed in *Totalitet som Kategori* (p. 3–20), but shall only make some remarks on the relation between valid and actual thinking. It is clear, I think, that since it is men who think the valid thoughts, the valid thoughts must also be actual thoughts, thoughts that are really thought — while conversely thoughts that are actually

thought are not always valid thoughts. Herewith the relation between psychology and epistemology is given. However high scientific thinking may raise itself, however many new means and forms of thought it may develop in order to solve its task, it must all [67] the same be psychologically possible, and psychology must, using the history of the sciences, investigate its possibility and its actual development. An anecdote which Kant somewhere tells on another occasion expresses the relation in a good image. An Indian, on a visit to a European, saw the cork spring out of a beer bottle and the froth foam out, and the Europeans were amused at his great astonishment at it. But then he said: "I am not astonished that it comes out, but at how you got it in!" Such is precisely psychology's position over against science, as over against art and religion. Psychology can as little bring forth science as it can bring forth art or religion. But it can and must set itself the task of investigating how these forms of spiritual culture are possible. History shows too that science arises out of the involuntary need for orientation in the world. If we are in doubt whether we stand before something real or not, or as to which road we are to strike into when we notice that we are astray, then we already have use for formal science, that is, for inferences, measurements and calculations, and for real science, that is, for the finding out of causal relations, processes of development and whole-formations, as regards the subject-matters before us. One of the most important differences between prescientific and scientific thinking is, as we have already had occasion to mention, that in the former involuntary expansion plays an essential part: one asks (if one asks at all) not "Why?" but "Why not?" — and lets things go as long as they will, whereas scientific thinking is born of doubt and tests every single step. Naturally a scientific genius applies the principles of strict thinking involuntarily; in the genius, as in instinct, need and [68] capacity are inseparably joined.

Kant already drew attention to the fact that "even the plain human understanding" thinks according to certain principles, and in the most recent time it has been energetically maintained, especially by ÉMILE MEYERSON with the support of the history of the sciences, that all science develops out of sound human understanding (*sens commun*),³ correcting

³The Danish expression "sund Menneskeforstand" is better than the German "der gemeine Verstand", the French "sens commun" and the English "common sense", inasmuch as in these last designations there lies something current, common and traditional, which is not always "sound". LÉON BRUNSCHVICG has aptly distinguished between "bon sens" as "fonction de discernement" and "sens commun" as "tradition de consentement universel", and finds this difference already drawn in Montaigne and Descartes. (*L'expérience humaine et la physique*. 1922, p. 574 f.).

its principles step by step where that is needed. There is not identity but analogy between science and sound human understanding. As Meyerson puts it: “The concepts of sound human understanding have been brought forth by a process which is indeed unconscious, but for the rest strictly analogous to the process by which the scientific theories are shaped” (*Identité et Réalité*² p. 393). The essential reason why science cannot stop at sound human understanding Meyerson finds in this, that although the latter can be led to seek and perhaps find lawfulness in what is lived through, it does not satisfy the need to set something lasting, something constant, in place of the more or less fleeting experiences; and he explains thereby the constant re-emergence of the atomic doctrine in the history of science. (*Op. cit.* p. 420–423.) But it would perhaps be just as correct to say that science does not find the lawfulness at which sound human understanding stops special, exact and comprehensive enough, and that the atomic theory keeps emerging [69] anew precisely because only this theory makes possible a quite exact representation of the most special lawfulness. — To this point of difference between the excellent French epistemologist and myself I shall later have occasion to return. —

By the following example the analogy between psychology and epistemology may perhaps be marked. Psychology draws attention to the fact that for a consciousness of time to develop, there must make itself felt in consciousness an opposition between something continuously subsisting and something changing. If one state were forgotten as soon as another state arose, consciousness of time would not be possible. A simple sensation of change, what we may call a sensation of time, presupposes a relatively fixed background against which the change can be noticed. In the simplest case this constant background will be formed by the feeling of life (the feeling answering to the given organic state), which can maintain itself while different, more peripheral experiences succeed one another. A representation of time proper (as distinct from the mere sensation of change) presupposes that certain experiences recur after an interval, so that a rhythm can be apprehended. Here a need or an impulse will be able to form the fixed background, and a rhythm of privation, striving, attainment (or disappointment) can be experienced. It is this practical triad that from the first carries the apprehension of time. Practically, the interest in the rhythm of time goes before the interest in the measurement of time, to which the history of religion and of magic testifies.⁴ An objective consciousness of

⁴HUBERT: *La représentation du temps dans la religion et dans la magie* (Hubert et Mauss: *Mélanges*

time develops through the observation of regularly changing natural phenomena (day and night, the seasons and so on) and forms, by the exact measurements astronomy [70] makes possible, the transition to the scientific apprehension of time. The alterations this apprehension of time has undergone in modern times I have had occasion to discuss in the treatise *Relation som Kategori* (p. 62–77). It is especially the question of the starting-point for the measurement of time that becomes decisive. As a consequence of the Copernican theory, Bruno maintained that every star must have its own time; and as a consequence of the electromagnetic theory of light, Einstein has maintained that the measurement of time depends on whether the system of co-ordinates laid at the base is relatively at rest or in motion. Even with the most exact and special scientific theory of time and its measurement carried through, the analogy to the primitive apprehension of time does not fall away, just as it has developed out of it through the constant work of centuries. —

It is with the concepts of number, degree and place in a similar way as with the apprehension of time. They too have developed, into the form in which science now works with them, out of qualitative observations, and the analogy with the initial stages has not fallen away. At all stages the principles for orientation in the world of subject-matters are at work.

14. The history of the sciences shows us remarkable examples of how certain fundamental thoughts emerge again and again, though in such different forms and with such different grounding that one is not seldom inclined to deny both the historical connection — which is nevertheless actually present — and the abiding significance they may have for continued research. There does after all make itself felt a continuity in the course of scientific development, and, connected with it, interesting analogies which point to a certain fundamental type for human thinking and for the demands it is to satisfy. [71]

a. The history of atomic theory shows us how through the ages two apparently opposite tendencies make themselves felt in human thinking: on the one side the need to find something that subsists in spite of all changes, and on the other the need to presuppose separate starting-points for the changes that experience demonstrates. Atomic theory arises through the co-operation of these two tendencies. It is due to the experience that one does not reach real understanding by stopping at the qualitative changes the senses show us. Democritus therefore declared that the sense-qualities had only apparent,

d'histoire des religions), p. 195.

not real existence. When one quality replaces another, something seems to arise and something to be lost; but when we apply “a finer organ of thought” than sensation, we reach a “genuine cognition” which shows us that the case stands quite otherwise: nothing arises or perishes, but what happens is that certain small parts group themselves in another way than before. These small parts are homogeneous, for otherwise they could not act upon one another. It is in the name of thought that Democritus maintains his atomic doctrine, and he was therefore set alongside Plato by Sextus Empiricus, since both maintained that only what can be thought (τὰ νοητά) can be true.⁵

Historically, Democritus’ atomic doctrine has exercised a great influence. His writings are, apart from some fragments, lost — one of the greatest losses within the literature of antiquity. But in the form Epicurus had given it, it was presented in brilliant fashion [72] in Lucretius’ famous didactic poem; it influenced Bruno in the Renaissance, and it was taken up again by Basso and Gassendi, from the last of whom Newton took it over (in his *Optics*). From Newton, John Dalton in turn took it over,⁶ in showing, following Lavoisier’s quantitative method, that the atom of an element has a constant weight, and that composite substances arise through combinations of different elements in determinate proportions by weight.

For the researchers of modern times there was this misgiving about Democritus’ and Newton’s atoms, that they were absolutely hard and indivisible. Then force would have to be lost at their collision. Huyghens and Leibniz therefore assumed, as a consequence of their hypothesis of the subsistence of force, that the atoms are elastic. Leibniz emphasized this very strongly against the Newtonian Clarke, who, like others of Newton’s disciples — e.g. Maclaurin — rejected the subsistence of force and found precisely in the absolutely hard atoms a decisive proof against it. Leibniz assumed that the force which the atom as a whole loses at collision is preserved in its parts, which are set vibrating (agitation) by the collision. Force is thus not lost, but is distributed over more points. It is as when large money is changed into small.⁷ On this utterance of Leibniz, HENRI POINCARÉ has remarked that one could not in any better way express the hypothesis that lies at the

⁵On an obscurity in Democritus’ doctrine of sensation and thought, see my treatise on Democritus and Plato. (Mindre Arbejder. III. p. 98).

⁶Cf. ROSCOE: *John Dalton*, p. 127–145.

⁷*Lettres entre Leibniz et Clarke*. (Leibnitii opera philosophica ed. Erdmann, p. 775).

base of the mechanical theory of heat.⁸ Through this modification of atomic theory the transition is formed to the apprehension of the most recent time, according to which atomicity is relative, every atom being a whole little world-system whose elements are called electrons. Democritus' "finer thinking" was thus not yet fine enough; but the science of modern times has developed the fineness in high degree — and why should thinking not be able to become finer still? [73]

ÉMILE MEYERSON has in his writings, and especially in an interesting discussion in the French philosophical society (*Bulletin de la Société française de philosophie* 3, mars 1910), exhibited with great expert knowledge the analogy between ancient and modern atomism. He will not say that Democritus and Lucretius have "priority" over modern chemists and physicists; even to call them forerunners would, he thinks, be to misapprehend the true spirit of the history of the sciences. And yet, he maintains, the setting together of ancient and modern atomic theory is not arbitrary; a real analogy takes place, in that the two theories spring from the same train of thought.

b. As it is with ancient atomism in relation to modern atomism, so it is on the whole with ancient thinking in relation to modern thinking. The opposition is clear, and often enough emphasized.⁹ For Plato, the most pronouncedly Greek thinker, it was science's task to form a system of concepts ("Ideas") under which all the subject-matters before us could be brought. In such a system one would have the key to all existence, he thinks. He sought in his own way the identity behind everything different and changing. The possibility of a classificatory system within which all the different subject-matters found their place under ever higher points of view of likeness was to him a testimony that the world of Ideas is the true world. In modern times the concept of law has been put in the place of the concept of Idea. Men do not flee from the differences and the changes, but find in the laws for the existence of subject-matters and for the changes that take place the identity which Plato found in the world of Ideas. Bacon and Bruno already interpret the Platonic Ideas as the laws of subject-matters, and Galilei's Platonic studies were of no small importance for his research. The principle of identity rules in the classificatory systems of the ancient thinkers as in the explanatory systems of modern times. In ancient science the qualities are sought reduced to identities through concepts of species and [74]

⁸See the quotation from Poincaré in LÉON BRUNSCHVIG: *L'expérience humaine et la causalité physique*, p. 227.

⁹See, e.g., *Den menneskelige Tanke*, p. 123–134.

of kinship. In modern science qualities are apprehended as identities by the help of determinations of quantity. Modern thinking too has its world of Ideas, namely in the circle of points of view from which it becomes possible not only to classify by the help of the principle of identity, but also — by the help of the same principle — to reach rational explanations. The world of Ideas has thus, as it were, sunk down into the world of experience; for it is no matter of course that the subject-matters observation offers should be able to make classificatory and explicative cognition possible. It might have been that existence was so constituted that there was no possibility either of classification or of explanation! But every victory, great or small, that science wins is a testimony to the contrary.

The analogy between ancient and modern science also lays itself open to view in this, that the difficulties which presented themselves for Plato's Ideas, and for which his genius and his love of truth gave him so clear an eye (see the first part of the dialogue *Parmenides*), make themselves felt in new forms for modern science's cognition of laws. For Plato the difficulty was this: when the world of Ideas is closed in itself, the one Idea being derivable from the other, and all Ideas finally from a highest Idea, how can there then subsist a world of subject-matters whose whole nature is not exhausted by the elements that find their place in the Ideas? In a corresponding way it stands as a constant [75] problem for modern science how the single subject-matters, in their concrete, individual peculiarity, should be derivable from the general laws, if these are to be science's last word or perhaps even to stand as the expression of existence's innermost essence. Here is a problem that makes itself felt both in the domain of natural science and in that of the human sciences. The problem sets itself in new forms; but it always follows thought on this road.

c. Within the research of modern times a significant analogy makes itself felt between the natural science of the seventeenth and of the nineteenth century. The seventeenth century founds the thought-life of modern times. The principal thoughts which later received their magnificent experimental confirmation come forward here for the first time. The principle of identity was boldly applied to the relation between cause and effect. Descartes and Spinoza declared that if there is in the effect anything other and more than there is in the cause, then full explanation has not been reached. Huyghens and Leibniz applied this thought in their assertion of the subsistence of force: neither can new force

arise at the transition from rest to motion, nor can force disappear at the transition from motion to rest. Against this thought the Newtonians CLARKE and MACLAURIN asserted that it conflicted with experience.¹⁰ But the thought was taken up again — as self-evident! — towards the middle of the nineteenth century by the researchers who at about the same time discovered the proposition of a relation of equivalence between the different special forces of nature, according to which at the transition from one form of energy to another energy neither arises nor disappears. A study of these researchers' works shows that they were led by the thought that there must be a relation of equivalence between cause [76] and effect. It stood for them as self-evident; but even the "self-evident" has its history, and this determinate self-evident thing did not exist before the seventeenth century. Fortunately the researchers in question (ROBERT MAYER, COLDING, JOULE, HELMHOLTZ) did not content themselves with a deduction from a "self-evident" proposition, but sought proofs from observations and experiments. A "leading thought" is, after all, not a thought that can without more ado be used as a major premise in a syllogism or as a presupposition for a deduction, but is a thought that can serve as a model ("Paradigm"), as a foundation for an analogy that can lead to the winning of cognition in new domains. The thought of the subsistence of force has significance in and for itself, namely in the domain of pure mechanics. But it was of course possible that its significance did not reach beyond this domain, so that it could not help towards an understanding of the mutual relations of the special forces of nature. This was the opinion of positivism's founder Auguste Comte, for whom the different branches of science were separated in principle from one another, so that a thought which can fit in the one domain had no significance for the other domains. The different branches of physics would, in his opinion, continue to stand separate from one another. He did indeed assume that in the long run it would not be possible for the human spirit to think strictly scientifically in one domain and unscientifically in others; there will involuntarily be a striving after unity in method and theory. But he ascribed no scientific significance to the analogy which here made itself felt; he derived it rather from the influence of habit. The founders of the proposition of the subsistence of energy, on the other hand, started out from the possibility that there could objectively be exhibited relations between the special forces of nature answering to the relation between motion [77] and rest in pure mechanics, and it was in reliance on this analogy that they went to their

¹⁰MACLAURIN: *Exposition des Découvertes de Newton*. Trad. franç. 1749, p. 88.

work. — We shall in what follows meet more examples of the significance of analogy where the relation between different sciences is concerned; but the analogy is not always so fruitful as in the case just discussed.

In my book *Ledende Tanker i det 19. Aarhundrede* I thought I could characterize this century as carrying through and verifying, in relation to the thoughts of the preceding centuries, at whose appearance the science of modern times came into being at all. In his fine and instructive presentation of the history of physics and chemistry in the nineteenth century, NIELS BJERRUM thinks that I have been too modest on natural science's behalf. To this I will say that it has by no means been my meaning to deny that the great thought-work performed within natural science in the nineteenth century has an independent and outstanding significance, and that much that is new and surprising came forward in the course of that century. But for a comparative historical consideration it is precisely new and surprising to see how thoughts which hitherto had come forward only in the abstract forms of epistemology and mechanics were now carried through in detail on the ground of experience and experiment. By "leading thoughts" must first and foremost be understood the directions in which thinking and research go. The direction is one thing, the thought-work another. The very introducing of a new direction does, however, require thought-work, and it may be a very difficult question to decide whether the founding of a wholly new direction in research (as when the principles of the new science were set [78] up in the seventeenth century) requires less or more thought-work than the carrying through of an already introduced direction in detail. Like all work, thought-work too is the overcoming of a resistance, and the resistance which thinking's first pioneers met was perhaps quite as great as that which the nineteenth century's researchers, who carried through their thoughts, met on their way.¹¹ The resistance has been of a somewhat different kind, and a sure measure for it we do not have. It was in any case not my task to investigate this point. I find the nineteenth century's greatness in the energetic faithfulness with which leading thoughts that had already seen the light beforehand were held fast and carried through.

d. Within the history of philosophy, HANS BRØCHNER has in his book *Bidrag til Opfattelse af Filosofiens historiske Udvikling* (1869) given interesting contributions to the

¹¹The literary fate which the first works of the four researchers mentioned above provisionally met testifies that here too there was resistance enough.

elucidation of analogy's significance for philosophical thinking. He finds a connection between the great periods in the history of philosophy, regarded as wholes, and the single attempts of thought within each of them, in that these attempts of thought now anticipate what comes forward in fully pronounced form only when the period has reached its high point, now continue to bear the period's stamp after it has properly closed. Brøchner assumes a mysterious influence of a period upon the single thinkers; but what lies before us is plainly analogies between attempts of thought which take place under approximately the same historical conditions, nearer to or further from those conditions' culmination.

The great periods which the history of philosophy presents are often introduced by the putting forward of views which are worked out more closely only later in the period. Thus in the oldest Greek philosophy there are anticipated the thoughts that first came to [79] full clarity in Plato. At the beginning of the Renaissance, attempts are made again and again at what Giordano Bruno first gives in systematic connection. In many respects Kant anticipates the thoughts of Romantic idealism. — On the other hand, thoughts that have been fostered within a period as a warranted expression of its special tasks can work hamperingly and limitingly on thinkers whose personal striving would in and for itself have had to carry them beyond the period's standpoint. The possibilities which presented themselves in Aristotle and Plotinus for a standpoint other than the classical Greek one were hampered by presuppositions common to all Greek philosophizing. But such a hampering influence from the past can often call forth more deeply going thoughts than would otherwise be possible. Thus Greek thinking has exercised a beneficial influence in the Middle Ages, and the thinking of the Middle Ages in modern times.

Brøchner regards the history of philosophy as a world closed in itself. The relations become more involved when it is taken into consideration that the way in which philosophizing is done is at any given time conditioned not only by the philosophizing of preceding times, but in high degree also by new experiences and by the work that takes place within other sciences. But even within its determinate limitation, Brøchner's fine and spirited consideration has its interest for the elucidation of analogy's appearance in human thinking.

Chapter IV

Analogy between Domains of Cognition

15. To the different domains of cognition, that is, to the different groups of sciences, there answer different groups of categories. Common to all sciences are the fundamental [80] categories: synthesis and relation, continuity and discontinuity, likeness and difference; and they are at the same time common to prescientific and to scientific thinking. To logic and mathematics there answer the formal categories: identity, negation, rationality. To the natural and the human sciences there answer the real categories: causality, totality, development. To the ethical sciences there answer the ideal categories: the value-concepts.

The analogy between the prescientific and the scientific consciousness I have already discussed. I now go over to exhibiting analogies between the formal sciences among themselves, between formal and real sciences, between real sciences among themselves, and finally between formal and real sciences on the one side and the ethical sciences on the other.

16. Logic and Arithmetic

Every science must itself, through analysis of its subject-matters and its methods, fix its necessary fundamental concepts in as simple and as fruitful a form as possible. But thereafter the question can be raised from the philosophical side whether the fundamental concepts thus set up do not in turn presuppose still simpler concepts, and, if they do, whether they can then without more ado be derived from these or must be set up as

new, more special concepts. A concept may very well presuppose another concept without being derivable from it. As regards the relation between logic and mathematics, the question becomes this: whether they lie on the same line, resting on the same foundation, so that mathematics signifies a straightforward continuation of logic; or whether mathematics comes forward with new fundamental concepts, between which and the purely logical fundamental concepts there can subsist only analogy, not identity.

- [81] Purely logical thinking would, as shown in the foregoing, stand powerless if all the subject-matters before us formed either a chaotic difference-series or an absolute identity-series. It is powerless both where there are no real likenesses at all and where there are no real differences at all. Even under these, logically regarded, desperate conditions, mathematical thinking can take one step more — and a step that can draw many steps after it. The differences or the identities can, namely, be counted; that is, a series can be formed between whose members there is identity of difference, a series that can always be continued according to the law of identity of difference, the same law by which the series begins, and it is indifferent whether the actually given series of differences or of identities is limited. The number-series is due to a construction which consists merely in the repetition of one and the same act of thought: every new member is found by positing a difference identical with the difference between earlier members in the series. The concept of number is built on a point of likeness between real differences (namely that they are objects for thought or attention) and a point of difference between real likenesses (namely that they are objects for different acts of thought). — The prescientific consciousness already attempts to form such series. The child “counts” its skittles (“one more, one more, one more” and so on) until it grows tired of it; the trouble is only with the summing up. The savage counts on his fingers and sums up by closing his hand. Even if no more subject-matters are given, the operation can be carried out by the help of self-created signs. Through such productive repetition the doctrine of number becomes something other and more than formal logic. From the principle of identity the possibility of always adding new members to those already given cannot be derived.
- [82] The principle of identity is applied logically to qualities as to quantities; but through productive repetition no new qualities arise; there is formed only a quantity-series or, as it can most simply be expressed, the pure concept of magnitude comes into being. Arithmetic’s fundamental concepts are thus more special than pure logic’s, and they can

as little be derived from these as a species-concept can be derived from a genus-concept at all. If one wished to say that a number is a general concept or a class-concept, like the other concepts logic operates with, there is all the same this important difference between number-classes (e.g. five-classes, six-classes and so on) and other classes (e.g. the class of horses): that whereas the definitions of the latter contain qualitatively different elements, the definition of a number-class contains only mutually identical elements.¹ The concept horse has for its content: “odd-toed hoofed animal with a specially developed middle toe and long metatarsus, with a complete set of teeth and with a bony ring about the eye”. The concept 6 has its content given in the series $1 + 1 + 1 + 1 + 1 + 1$, which arises through the repetition of an identity of difference — a repetition we can ourselves carry out without awaiting observation. For logic the number-series is only an example of a peculiar kind of series-formation, an identically varying difference-series or also a progressive difference-series (according as the talk is of ordinal or of cardinal numbers). It is indifferent for logic how such a series comes into being; it takes it as a fact that it can be formed. But what for logic is a single example is for arithmetic of fundamental significance. The number-series is grounded on a postulate, in that the possibility of a construction such as the one described is asserted. It is through this postulate that we come from logic over into arithmetic. And the same process which leads to forming [83] the number-series leads to continuing it through negative, fractional, irrational and imaginary numbers.

This postulate, which contains what may be called “mathematical induction” or “the law of repetition” (*loi de récurrence*), was applied in purely mathematical form by Maurolyco (16th cent.) and by Pascal (17th cent.).² JOHN LOCKE applies it as a foundation for the concepts of number, space and time (*Essay* II, 12, 8 and 13, 1–2). As characteristic of mathematics in opposition to logic it has been apprehended by Leibniz, Kant, Boole and Poincaré.³ Plato’s doctrine of mathematics as lying between pure logic (“dialectic”) and the world of observations (*Staten* p. 511 A. D; 525 A; 527 B) is thereby confirmed.

In the most recent time HILBERT has declared himself not only against those who,

¹Cf. already PLATO: *Staten* VI, p. 526 A, where it is said of the arithmetical units: ἴσον τε ἕκαστον πᾶν παντὶ καὶ οὐδὲ σμικρὸν διαφέρειν, μῶριον τε ἔχον ἐν ἑαυτῷ οὐδέν.

²*Œuvres de Pascal*, éd. Brunschvicg. III, p. 456; VIII, p. 363 (with Pierre Boutroux’s notes).

³Cf. *Totalitet som Kategori*, p. 33–39.

like Frege and others,⁴ apprehend the doctrine of number as a part of logic, regarding numbers as species-concepts, but also against those who, like Poincaré and others, build it on the capacity to repeat an act of thought. This last conception Hilbert designates as mystical. In supposed opposition to it he maintains⁵ that one must begin with “certain discrete objects, given in an intuitable way prior to all thinking”. Such objects we have in the signs, the doctrine of number’s proper objects: “Am Anfang ist das Zeichen!” The sign 1 is a number. And a sign which begins with 1 and ends with 1, in such a way that [84] in the interval after each 1 there always comes a + and after each + a 1, is also a number. Thus we get the series $1 + 1 + 1 + 1 \dots$. In order to be able more easily to communicate with others, we then form the sign 2 for $1 + 1$, 3 for $1 + 1 + 1$ and so on.

I cannot see otherwise than that this arrangement leads back to the conception which has been maintained by Poincaré and his mathematical and philosophical predecessors. A sign and a series of signs come into being, after all, only through acts of attention or of thought, and a series such as the one set up presupposes the possibility that such an act can be repeated in one and the same way (which naturally, psychologically regarded, is only approximately possible). There is nothing mystical in the fact that we can repeat our thoughts or, if one will, our signs in such a way that the series is constantly continued according to the same rule which led to the setting up of the two first members.

In this, however (and it is in this connection the most important thing), Hilbert agrees with the other group of researchers: that mathematics cannot be derived from logic. A sign-language of the kind he introduces logic has no use for.

There rules, then, no identity between arithmetic and logic. But analogy does rule. Number can stand to number as ground to consequent, and Hilbert accordingly has use for the logical sign $\rightarrow$ as an expression for the relation between ground and consequent. (It is the same sign that I used in *Den menneskelige Tanke* p. 168 to denote the relation between the members in a partial identity-series.) The purely logical relations are found again in the mathematical domain, and that in so uniquely clear and exact a way that the best and most fruitful examples of logical inferring lie before us here. But they are all the same only examples among other examples. And within the doctrine of number there is [85] analogy between algebra and arithmetic, in that in algebra one fastens attention only

⁴So B. RUSSELL: *Introduction to mathematical philosophy*. Chap. 2.

⁵*Neubegründung der Mathematik*. (Abhandlungen des mathematischen Seminars der hamburgischen Universität. I, 2 (1922), p. 161–163.

on the way in which the magnitudes are combined, without knowing the magnitudes themselves. Thus $(a + b) c^2$ expresses all cases in which a magnitude is determined by the product of the sum of two magnitudes and the square of a third magnitude, indifferently as to what values a , b and c have in the special cases.⁶ We find again here the logical relation between a concept and the examples for which it holds. Every example must be substitutable for another everywhere, if the concept is to hold. In the doctrine of number this substitution is unconditional; with concepts of qualitative subject-matters it holds in each single case only from one determinate point of view.

The relation between logic and arithmetic becomes analogy; on the foundation that comes into being through the construction of the number-series by productive repetition, a new and more perfect application of the logical relation between ground and consequent is possible; but that construction cannot be derived from pure logic as such. There is given a new contribution in cognition's development with the arising of the concept of number.

17. Arithmetic and Geometry

Not only the doctrine of number, but geometry too rests on the postulate of the possibility of repeating one and the same act of thought. Whatever an act of thought's content or object may be, I can always add a new act of thought to the earlier ones. I can think of more horses than those I have seen, and more "signs" than those I have already written down, more line-segments than those I have already drawn. But on closer investigation men have all the same been led to draw an ever sharper distinction between the doctrine of number and the doctrine of other subject-matters, whether these are geometrical figures [86] or concrete objects. What is peculiar to number is its discrete nature. It is determined merely by being a unit alongside other units, or by consisting of units that have this property. But in the subject-matters that are counted there are always other properties besides that of being countable. That is already the case with geometrical lines. In and for itself one could, instead of expressing units by vertical strokes ($1 + 1 + 1 + 1 \dots$), use horizontal strokes ($- - - - \dots$); and when then the one immediately continued the other, a straight line would be constructed. This arises then, to that extent, by the

⁶PIERRE BOUTROUX: *L'idéal scientifique des mathématiciens*. 1920, p. 84.

same “Synthesis des Gleichartigen” as the number-series. KANT has thought of the matter in precisely this way when he says: “Wir können uns keine Linie denken, ohne sie im Gedanken zu ziehen”, and when he maintains that we “im Ziehen der geraden Linie bloss auf die Synthesis des Mannigfaltigen Acht haben”. (“Kritik der reinen Vernunft”² p. 154.) There does after all take place an addition at every moment while we draw the line, just as every time we add a unit in the number-series. But the great question is whether we have the right to say that the constant additions, while we draw the line, are equal. With the number-series this question did not arise, and for the simple reason that each single member is regarded merely under the point of view of an addition — solely as something that is added, without there being any talk of extension or of other properties.

This difference in principle between the doctrine of number and geometry has worked its way forward successively. Greek mathematics had its foundation in geometry. Relations of number were made intuitable by relations of space — the product of two numbers, e.g., by the rectangle of two lines representing the factors. With Descartes came a turning-
 [87] point. Not only were figures now conversely expressed by numbers and equations stating the relation between the figures’ elements, but geometrical problems were reduced to algebraic problems. There was, as ZEUTHEN⁷ puts it, a complete reversal of the traditional relation between algebra and geometry, and thereby is marked a reform in the whole of mathematics, which hitherto had had its acknowledged foundation in geometry but from now on got it in algebra. A step further went MAXWELL (1855),⁸ when he declared that all scientific applications of mathematics rest on relations between the laws of physical magnitudes and the laws of the whole numbers. When BOLTZMANN, in a note to his translation of this passage in MAXWELL, says that so far as he knows the standpoint that measurements of magnitudes of space and time by numbers rest merely on analogy with the relations subsisting between the numbers has not since been taken up, this no longer holds. Several researchers have continued MAXWELL’s train of thought. Thus ERNST MACH (who in general has drawn attention to MAXWELL’s epistemological way of looking at things) lays great weight on the limits of measurement’s exactness. “The coincidence of a measuring-rod and the subject-matter that is to be measured cannot”, he says,⁹ “be established with unlimited exactness”. And the step has been taken right

⁷ *Mathematikens Historie*. II, p. 285.

⁸ *Über Faraday’s Kraftlinien*. Übers. von BOLTZMANN (1895). Einleitung p. 4.

⁹ *Erkenntnis und Irrtum*. (1905), p. 328 f.

out by HJELMSLEV in his presentation of analytic geometry.¹⁰ HJELMSLEV distinguishes sharply between the geometry of reality, which deals with things, and abstract geometry, which deals with numbers. In a chapter on the theory of geometrical measurement he shows that no measurement of a line-segment presented in a drawing can be wholly exact; not one but several results are obtained, between which a choice must be made if [88] the line-segment is to be “fixed”. Drawings are constantly necessary for guidance and description, but they can be apprehended only as symbols of the abstract lines which can be expressed by determinate numbers.

ZEUTHEN¹¹ regarded, in the last writing we had from his hand, the ideal or abstract figures — whose concepts are formed by detachment from what sensation shows — as symbols, just as the letters and operation-signs of modern mathematics may be said to be. Such ideal or abstract figures can now, according to HJELMSLEV, be expressed only in numbers. In an interesting way the two Danish mathematicians have thus each seized his own side of the correlation which the concept of symbol contains. When *A* is a symbol of *B*, *B* is namely also a symbol of *A*. HJELMSLEV, whose conception of geometry is most nearly to be called realistic, apprehends the real geometrical forms as symbols of the abstract forms, which according to him are expressed only by numbers. He can say with GOETHE: “Alles Vergängliche ist nur ein Gleichnis!” But ZEUTHEN, whose conception of geometry is most nearly to be called idealistic, since he maintains the necessity of ideal definitions in geometry, conversely regards these ideals as symbols of the real geometrical relations. —

There lies here an extraordinarily interesting example of how analogy can be the only way in which the relation between different domains of cognition can be marked. And here, as at other points, the development of thinking has led to men’s finding analogy where they formerly found identity.

18. Formal and Real Science

Analogy will now show itself to be the right expression not only when the relation [89] between the different formal sciences (logic, arithmetic, geometry) among themselves is to be determined, but also when the relation between formal science and real science is

¹⁰*Elementær Geometri*. Tredie Bog. 1921, p. 71; 91; 180 f.

¹¹*Hvorledes Matematikken i Tidsrummet mellem Platon og Euklid blev rationel Videnskab* (1917), p. 9.

to be determined. — We keep provisionally to natural science as the representative of real science, and go over later to discussing the relation between natural science and the human sciences.

It is the relation between the concepts of ground and cause that is here in question. On this point lies KANT's great deed. He has set analogy as the expression for the relation between ground and cause, in place of the presupposition of these concepts' identity from which dogmatic philosophers and investigators of nature had started out. The relation between cause and effect signifies, according to KANT, only that the succession in time of two subject-matters (events) is analogous to the logical-mathematical relation between ground and consequent. Real science works without interruption to demonstrate that every change that occurs in the world, everything new that appears for our observation, stands to preceding changes as an inference stands to its premisses, that is, as consequent to ground. All mystery in the causal relation thereby falls away. We know nothing of this relation except that it is at once a relation of ground and a relation of time. Schematically we have here an analogy between two series: a partial identity-series ($A \rightarrow B \rightarrow C \rightarrow D \dots$), in which each member stands to the following as ground to consequent; and an indeterminately varying difference-series ($a_{\mu} b_{\nu} c_{\pi} d \dots$), where the members' order of succession is merely chronologically determined. From a member in the first series one can infer to a determinate member in the second series, and thereby, within the [90] second series, from one member to a following one, even if one thereby gets outside the members actually before us. A cause is a subject-matter which appears for our observation at a certain point of time, and from which it can be inferred that a certain other subject-matter will appear at a later point of time. It was through his study of Newton's physics, and under the influence of the sting Hume's criticism of the concept of cause had left behind, that KANT arrived at this conception.¹² By virtue of the analogy between a logical-mathematical relation and a relation of succession, experience becomes something other and more than a collecting of single observations. This new concept of experience KANT sets in opposition to Hume's conception of experience as a sum of isolated, mutually independent observations — that is, as a chaotic difference-series, Spinoza's *experientia vaga*. But his exposition stands in high degree in need of closer

¹²The clearest passages in KANT are *Kritik der reinen Vernunft*¹ p. 112 and *Prolegomena*, § 30.

determination and simplification¹³ before it can be taken up as a member in our own time's epistemology.

a. The causal relation itself KANT discusses in the section of the "Kritik der reinen Vernunft" headed "the Analogies of Experience". But two sections go before it, called "the Axioms of Intuition" and "the Anticipations of Perception", and here KANT does not seem to think he has use for the concept of analogy. — "The Axioms of Intuition" lays it down that all phenomena have extension in space or time or both. What KANT here discusses is nothing but an application of what has above in this treatise been called an identical quality-series, which can be constructed by the logical-mathematical way, and whereby the concepts of number, place and time acquire an ideal form, since there is [91] identity of difference between the members of the series — between the numbers, the places and the moments each for itself. But the expression analogy fits better here than the expression axiom. The meaning is that every subject-matter must be referable to, or able to enter as a determinate member in, one of these series. In spite of all special differences of quality, these series hold for everything about which we are to have strict experience. But between the series of the special differences of quality and these series there is only analogy, not identity. — And it stands in a corresponding way with "the Anticipations of Perception". Here the talk is of the series of degrees. Every subject-matter that is to be able to be an object for strict experience must be able to enter as a member in a series within which each member, as regards intensity, always has the same difference from its predecessor — that is, a progressive difference-series, like the series of numbers, of places and of times. Since the actual subject-matters present many other properties than intensity, the relation between the series of degrees and the series of actual subject-matters can only be analogy. But it is apt when KANT here speaks of the anticipations of perception; only he ought to have used this expression of all his epistemological principles. Analogy can lead to no more than a grounded expectation (i.e. a hypothesis), which then possibly is confirmed by new observations and experiments.

b. KANT sets up three analogies of experience. — The most important is the second, which concerns the causal relation in its generality. Here it is shown, as already discussed, that a causal relation is a relation of succession between two subject-matters which

¹³In the third edition of *Den nyere Filosofis Historie* I have already (III, p. 278 f.) given the grounds for the necessity of a simplification of KANT's exposition. But there I kept mostly to the purely formal side.

presents an analogy with the relation between ground and consequent. — The first [92] analogy maintains that in a causal relation no increase or decrease occurs in what subsists. It is “the Analogy of Substance”. If it were to be set up at all, it ought to have been placed after the general causal proposition, which it plainly enough presupposes. What it demands is that the relation between ground and consequent which the causal proposition presupposes shall be reversible, so that the consequent can become ground and the ground consequent — to which it answers that what was first effect can be cause of what was before cause, and conversely, so that there is a relation of equivalence between cause and effect. But such a proposition can be set up only as anticipation and as ideal. The history of science shows that everywhere it is possible, men seek to exhibit such a relation of equivalence. Thereby one approaches a pure relation of identity to the degree that is possible when one has to do with qualitatively different subject-matters; and one then at the same time abstracts from the difference of time which in each single case is present and may be of decisive importance for the direction in which the following changes take place. But not all inferences can be reversed, any more than all predicates can be made subjects for their earlier subjects as predicates. If KANT’s “analogy of substance” holds, it is in any case a particularly happy case of the causal analogy. And it is an unfortunate expression that KANT uses when he here speaks of substance as a special category. He does not take the word substance in the old, metaphysical sense, but in a relative sense, as he himself expressly declares (*Kritik der reinen Vernunft*¹, p. 525): “Was in der Erscheinung Substanz heisst, ist nicht absolutes Subjekt, sondern beharrliches Bild der Sinnlichkeit und nichts als Anschauung, in der überall nichts Unbedingtes angetroffen [93] wird”. The concept of substance in the old, absolute sense is a dying category.¹⁴ — As regards the third analogy, which maintains that everything simultaneous must stand in reciprocal action, the matter is quite simple, for reciprocal action presupposes causality. If we are to understand, starting from a given subject-matter, the simultaneous existence of another subject-matter, there must through more or fewer intermediate members be a relation of causality between them or between their causes. We must, starting from the one subject-matter, by the requisite orientation come upon the other.

c. All three analogies together assert, according to KANT, that all subject-matters lie in

¹⁴Cf. *Den menneskelige Tanke*, p. 217 f. — On Spinoza’s concept of substance see *Spinoza’s Ethica*, p. 6; 19–20. Cf. besides above, § 7 at the end.

one nature. By “nature” KANT understands precisely “the connection of subject-matters, as regards their existence, according to necessary rules, that is, according to laws”. (*Kritik der reinen Vernunft*¹, p. 216.) KANT’s concept of nature, which coincides with the strict concept of experience, thus maintains analogy between the series of successive and simultaneous subject-matters and series of ground and consequent and of numbers, times, places and degrees. If experience is to be possible, the subject-matters must be capable of being ordered in accordance with such series.

d. While KANT carefully constructs the “schematisms” — that is, the progressive difference-series whose application to the subject-matters before us is to make experience in the strict sense possible — and while he carefully distinguishes between the analogy which thereby plays a part and identity, he has not seen that only an advancing, approximating application of this analogy is possible. His concept of nature or of strict experience is rightly formed, but he did not see that “experience” in this sense is an ideal, a great hypothesis which properly can never be fully confirmed. It holds here too that a hypothesis’ success does not prove its truth. The great question (which Hume raised, [94] and which KANT thought he had answered by his definition of experience) is whether all subject-matters that can be apprehended now and in the future permit an application of those ideal series. KANT’s “analogies of experience” formulate in a way of genius the great hypothesis on which natural science has worked for three hundred years, and which has again and again shown itself fruitful. And even if he did not see the hypothetical character of his concept of experience, it was all the same a great achievement that he saw and formulated so clearly the principle that lies at the base here. Therefore he still marks a principal point and a central point in the epistemology of modern times.

e. It was no wonder that a conception such as KANT’s could not at once prevail. On the contrary, the connection between formal and real cognition which he had sought to exhibit was dissolved. The tendency to reach absolute identity as at once the highest form of thought and the highest result of thought, the wish once for all to get beyond everything chaotic and manifold to rest in the thought of something that does not again point beyond itself, was — especially under the influence of poetic and religious currents in the spiritual life of the time — too strong for men to be able to content themselves with KANT’s apparently modest analogies. The realities were to be wholly taken up into the “Idea”, into the pure form of thought. In German Romantic speculation this tendency

comes forward. GOETHE and SCHELLING maintained that the explanation of the qualities was to be found within the world of qualities itself, certain qualities being laid at the base as “primal phenomena”. HEGEL attempted to derive the world of qualities and of time from the world of pure logical identities — an attempt which at once, by its confidence in the power of pure thought, bore the stamp of Romanticism, and pointed beyond Romanticism, [95] since HEGEL acknowledged that full understanding cannot be reached by stopping at the qualities.¹⁵ Every science that will give absolute solutions must end in an attempt like HEGEL’s. If those are right who maintain natural science to be the absolute science, they must make the attempt to derive the qualitative differences from the logical-mathematical principles by whose help the world of qualities becomes intelligible to us. If one will not acknowledge that strict real science reaches its goal through exact analogies between quality-series and mathematical-logical series, one must make HEGEL’s attempt over again and crown modern natural science’s striving to explain all changes as motions whose laws can be formulated as equations, with an attempt to derive the differences of quality from the identities lying at the base in the equations. Then there would be given a proof that natural science had solved the riddle of the world. The “parfait retour” which Leibniz demanded would have been reached. Such an ideal surely stands in the background of many investigators of nature’s train of thought. If one asks what then becomes of the qualities, which are after all the constant and immediately given, the answer runs that they are only “anthropomorphisms” — the same answer the theologians give when it is asked how the psychological expressions used of the Godhead’s essence and working are properly to be understood. But in both cases the “anthropomorphism” sets a new problem. Qualities such as colour, sound and so on, and such as thinking and willing, simply are subject-matters that lie before our cognition, and the question is whether they can ever be absorbed, wholly taken up into rational knowledge (whether [96] this comes forward as a series of equations or as a series of dialectical transitions). We can, as the history of natural science shows in brilliant and unique fashion, build up a world of thoughts that can guide us in the world of “anthropomorphisms” in which we live. And this guidance natural science gives us, if the consideration set forth in the foregoing is correct, by the help of exactly carried through analogies. — Whether theology can

¹⁵In his spirited characterization of Hegel’s philosophy of nature, ÉMILE MEYERSON lays at the base a point of view similar to the one applied here. Cf. *De l’Explication dans les Sciences*, II, p. 78: “Hegel a réellement tenté une entreprise qui constitue un postulat éternel de l’esprit humain”.

give us a corresponding guidance I leave here undecided. Attempts at a strict carrying through of the analogies applied lead in quite another direction than positive theology.¹⁶

The impossibility of deriving differences and changes of quality from the equations that mark the formal foundation of natural science's exact analogies — and in which there is a certain inclination to see the true reality — is seen already from this, that the formation of the identical quality-series (the series of numbers, times, places and degrees) takes place by an analysing purging-out of the empirical quality-series. From the concrete wholes before us one takes out the formal elements which permit strict logical-mathematical treatment. In strict science's exact results it is always only a part of reality that is expressed, even if it is the part of reality which by the way of analogy can lead to the understanding and the mastering of the other parts or sides of reality. It is always nature that is explained by nature itself, in that the most transparent relations are laid at the base for the interpretation of other relations. Or, to use KANT's definition of "nature": that which makes nature "nature" is used to interpret the parts of nature which do not yet satisfy the strict concept of nature. In contrast to the primitive mentality, [97] the basis of interpretation is now definitely brought out, and it is no longer any foreign authority that is consciously or unconsciously invoked. But just as little as cosmology (metaphysics), which always wishes to explain the whole of existence from a part of it (from "matter", from "spirit" and so on), has ever been able to reach a rational conclusion, just as little will any science be able to do so.

In the most recent time the Marburg school has apprehended the elimination of the qualitative elements as the goal of scientific cognition, regarding only quantitative science as true science. Its attempt is a companion piece to Hegel's, only that a system of equations takes the place of speculative dialectic.¹⁷ In my treatise *Totalitet som Kategori* (p. 46–48) I have had occasion to state my position towards this tendency, which is one of the most energetic and most significant that have developed since the taking up again of the problem of cognition in the last half of the nineteenth century.

With other epistemologists Kant's thought has come more into its own. Positivism's founder AUGUSTE COMTE laid great weight on the difference between the different sciences, which he ordered in a series beginning with mathematics and in whose following

¹⁶Cf. my *Religionsfilosofi*, § 16–25.

¹⁷Cf. GUSTAV KÖRBER: *Der Marburger Logismus und sein Verhältnis zu Hegel*. (Archiv für die Geschichte der Philosophie. 1917).

members special and complex qualities played a greater and greater part. The last member in this series is sociology. Between the different sciences there can be only analogy, not identity; and COMTE even misapprehended, as we have had occasion to see, such reductions as were scientifically warranted. But his classification of the sciences, [98] which advances from the simple to the complex, follows an ordering similar to the one applied shortly afterwards by RENOUVIER in the doctrine of categories,¹⁸ and which is also followed in my own exposition. It holds for categories as for sciences that the more complex cannot be derived from the more elementary, but that there may all the same subsist instructive analogies between them. With full consciousness and great clarity the point of view of analogy was maintained towards the middle of the nineteenth century by CLERK MAXWELL, as we have already had occasion to see as regards the relation between arithmetic and geometry. Analogy expresses in general, for MAXWELL, the relation between formal and real science. Attaching himself to him, another philosophizing investigator of nature worked, ERNST MACH, whose epistemology is built above all on an independent study of the history of natural science.¹⁹ Of more recent researchers must especially be named ÉMILE MEYERSON in his works already several times discussed. When PIERRE BOUTROUX²⁰ declares that mathematics is a method, not an objective science, a similar conception of the relation between formal and real science seems to lie at the base.

With outstanding investigators of nature, even where they do not, like MAXWELL and MACH, go more directly into epistemological questions, one will all the same often find utterances that testify to full clarity concerning the relation between data and their rational interpretation. The word analogy is perhaps not named, but the expressions used point decidedly in the direction of the concept of analogy. Already in the title of NIELS BOHR's treatise *Atomernes Bygning og Stoffernes fysiske og kemiske Egenskaber* (1922) this [99] concept is really intimated. The physical and chemical properties of substances are the given. The qualitative differences they present — the atomic weights of the elements, their spectra and so on — can be ordered in series, to which there answer, in the theory of the atom's structure, positions and transitions of the electrons within the atom. What interests epistemology, apart from the results significant for objective science, is the

¹⁸Cf. *Relation som Kategori*, p. 13.

¹⁹MAXWELL and MACH I have attempted to characterize in *Moderne Filosofer*, p. 82–92.

²⁰*L'idéal scientifique des Mathématiciens*, p. 152; 243.

way in which these results are expressed. There are always intimated in the expressions analogies, not relations of identity — a “signifies”, not an “is”, between the quality-series which forms the starting-point and the quantity-series which leads to a determinate thesis about atomic structure. Thus there is talk (p. 1) of “elucidation”, (p. 33) of “explanation, or rather understanding”, (p. 36) of “interpretation”, and on p. 45 it is said: “as the spectrum shows, and as the atomic model makes intelligible”. The construction of the atom’s inner structure, and of what goes on within it, is guided step by step by observation of changes in the qualitative properties of substances. — It is clear that if the elements did not allow themselves to be ordered in series where the members followed one another according to a certain law — if they thus formed a chaotic difference-series and not a progressive difference-series — no rational analogy could be set up.

f. If it is right that the relation between formal and real sciences is a relation of analogy, then there lies in this that advances in the domain of the real sciences cannot be won by deduction from the results of the formal sciences. As a method an analogy can only be a leading thought, an anticipation, not a proof. It can lead to the putting of questions, and a well put question will in a scientific mind lead on to closer investigations.

The fundamental concepts of our cognition themselves, the categories, lay themselves [100] open to view most clearly in the questions that are put.²¹ In the leading thoughts which lead to questions and investigations there shows itself above all the continuity in the history of science. A new circle of such leading thoughts was formed at the founding of modern science nearly 350 years ago, and these thoughts have preserved their force to this day — especially after the nineteenth century had worked with so great energy to put experiences in the place of analogies. As already discussed in another connection, this strikes the eye particularly in the study of the treatises in which ROBERT MAYER, JOULE, COLDING and HELMHOLTZ, at about the same time but independently of one another, founded the modern doctrine of the subsistence of energy. They all declared that they at bottom regarded it as a matter of course that there must be equivalence between cause and effect — which fortunately did not keep them from investigating the relation through observation and experiment. But the matter of course they assumed has not been acknowledged at all times. It is no older than the seventeenth century, whose philosophers and investigators of nature maintained that in the effect nothing new could

²¹See on this *Den menneskelige Tanke*, p. 137 f.

be added and nothing lost of what was in the cause. HUYGHENS and LEIBNIZ maintained this proposition in mechanics, after DESCARTES and SPINOZA had maintained it from what we should call an epistemological point of view. The proposition was, in the manner of the time, set up purely dogmatically. SPINOZA does indeed ground it by saying that if there were anything other and more in the effect than in the cause, this more would have no cause and would thus stand unexplained. But who says that everything must be
 [101] explicable? What was to be shown was precisely that explanation was here possible, and the great thing about the discoverers of the energy proposition is that they prepared the way for it.

But even if analogy has significance only as a foundation for questioning and seeking, or as a heuristic principle, it is naturally of importance that it be grasped in its determinateness and its limitation. We have seen that this was still wanting in KANT when he thought he could set up a proposition about “the subsistence of substance” even before he had set up the causal proposition. His discovery of genius was that between the concepts of ground and cause there was only analogy, not absolute identity. But as regards the concept of ground itself (or the relation between ground and consequent), he did not distinguish between inferences that can be reversed (so that there is equivalence between ground and consequent) and inferences that cannot be reversed. All real science stands and falls with the causal proposition, according to which one subject-matter stands to another subject-matter as consequent to ground; but that ground and consequent (cause and effect) should be interchangeable is a more special assumption which cannot be derived from the general causal proposition, even though it may be said to formulate the ideal for all real science. There can be science even if science’s highest ideal is perhaps never reached. — With ROBERT MAYER a similar lack of clarity makes itself felt. A friend of his youth, who stood in lively communication of thought with him before his epoch-making treatise was composed (1841), says of him: “Though he was far from being a schoolman philosopher, a quite independent reflection in the domain of logic and metaphysics on the essence of causality had perhaps quite as great a share in his
 [102] discovery as exact investigation of nature”.²² But MAYER thought, as is seen from his foundational treatise *Bemerkungen über die Kräfte der unbelebten Natur*,²³ that his new

²²GUSTAV RÜMELIN: *Erinnerungen an Robert Mayer* (Reden und Aufsätze. Neue Folge. 1881, p. 383).

²³Reprinted in *Die Mechanik der Wärme*. Herausg. von Weyrauch, p. 29.

proposition followed “with necessity” from the proposition *causa æquat effectum*, and that there was therefore equivalence between motion and heat. MAYER did not know that the corrective to his proposition had already been given in a treatise by CARNOT (1824), which showed that the transition from heat to motion does not go on as easily as the transition from motion to heat. And he did not see clearly that in the transition from logic and mechanics to special physics there can be no talk of deduction, but only of an analogical reflection of the abstract ideals in the concrete relations between qualities. — Analogy’s methodological significance naturally does not fall away because it must in many ways be reshaped and limited by the very work that is done under its guidance. —

g. Of analogies with heuristic significance within one and the same domain, those are especially of epistemological interest in which one starts out from an ordering within a certain totality and investigates whether a similar ordering is not found again within other totalities. For Galilei, thus, the discovery of Jupiter’s moons contained an analogy for the elucidation of the sun’s relation to the planets (the earth included), and so for the strengthening of the Copernican conception. And in the most recent time the analogy between the inner structure of the solar system and of the atom has worked to make things intuitable and to carry them forward. Such examples lead us over from the concept of causality to the concept of totality. There lies in the nature of thinking a need to form wholes. Every inference is a logical whole, and at the base of the causal need there lies likewise a need to abolish the single subject-matter’s isolation by bringing it into as firm a connection as possible with other subject-matters. But from this no deduction leads to special results. The application of the concept of totality rests, like that of the concept of cause, on an analogy which may be of great importance as a leading thought but has no probative power. As I have had occasion to say in another treatise (*Totalitet som Kategori*, p. 51), the concept of totality can never be used as an explanation, as happens in vitalistic, spiritualistic and mystical-sociological theories. Like all categories, this concept is a form for thought-work and at the same time an ideal which is sought to be found again and carried through in the particular. Where observation immediately presents subject-matters with a character of wholeness — an atom, a heavenly body, an organism, a personality, a society — there we are set just so many tasks: first for description and classification, then for the investigation of their arising and of the conditions of their [103]

subsistence.²⁴ It is an analytic work on the foundation of what observation can present that is here required. As LEIBNIZ has said, it is the law for the connection of the elements that constitutes a being's peculiar wholeness, and it is such a law that the analytic work is to seek to find.

As it stands with the concept of totality, so it stands also with the concept of development. This concept too is a form of thought which guides research both involuntarily and with full consciousness. Its methodological significance shows itself in this, that it contains an invitation to attempt to construct a series in which each member conditions the following and is conditioned by the preceding, and that in such a way that in later members a greater manifold of elements is always connected in a more intimate way.

[104] It becomes a regularly varying difference-series which cannot be reversed. (Cf. Den menneskelige Tanke, p. 233 f.) As regards the development of the single organism or the single organ, the doctrine of epigenesis founded by CASPAR FRIEDRICH WOLFF maintains that there can be only analogy between the different stages, whereas earlier theories found identity here, regarding the later stages as completely preformed in the earlier. The same thought as in WOLFF appeared at about the same time in ROUSSEAU, in his psychological-pedagogical theory, which maintained that the difference between the child and the adult was not merely a difference in the dimensions. In the nineteenth century the concept of development was also attempted to be carried through for the organic species, and men sought to exhibit a continuity in the processes of development, whereby the analogy between the different stages acquired a real foundation. CUVIER was still content that the analogy had its ground in the Creator's imagination, which after every destruction of organic life caused by a geological catastrophe brought forth new, analogous forms. In GOETHE and the Romantic investigators of nature, metamorphosis and development were only figurative expressions for the analogy of the forms; they assumed displacements in nature's systematic imagination, but not any real descent. Many of DARWIN's opponents, especially anatomists and morphologists, stood at such a standpoint. When modern strict critics of Darwinism sometimes declare that they do not understand the standpoint those morphologists took, it is a sign that the analogy between the stages of development has to such a degree shown itself in its real significance that researchers of the present cannot enter into their own subject's past. Nothing better

²⁴Cf. more closely *Totalitet som Kategori*, p. 52–66.

shows the importance of the study of the history of science with special regard to the fundamental forms within which research moves. Those who are led by certain thoughts [105] do not always discover the leading-string, and the consequence of that can be not only a lack of historical understanding, but also a dogmatic confidence in the forms of their own time.

Within biology a sharp distinction is drawn, especially since OWEN's time, between analogy and homology,²⁵ in such a way that the former rests on an agreement in function and is thus of a physiological kind, while the latter rests on agreement in structure and is thus of an anatomical kind. The lungs of mammals and the swim-bladders of fishes are homologous, not analogous; the lungs of mammals and the gills of fishes, on the other hand, are analogous, not homologous. There can be a dispute between anatomists and physiologists whether analogy or homology is of the greater importance. In part the famous dispute between CUVIER and ST. HILAIRE was of this kind. CUVIER thought essentially of unity in structure (*unité de composition*), while ST. HILAIRE understood by unity analogy.²⁶ Yet it must be remarked that epistemologically likeness of form, likeness in structure, is often to be called analogy, since it is more the relation between the parts than qualitative likeness at single points that is in question. According to the doctrine of epigenesis there is, as already remarked, only analogy between different stages of development of the same organ, even if the development is seen to have gone continuously from the one form over to the other; and according to CUVIER's own account, ST. HILAIRE thought essentially of analogies of structure in different species. Seen from this side it was thus a dispute about likeness of form or likeness of relation. As I have remarked in my *Psykologi*, likeness of form is a transitional form between qualitative [106] likeness and likeness of relation.

h. As regards the understanding of organic life, the possibility must, as already touched on, be excluded that the concept of organism should itself contain an explanation. Concepts such as life and organism stand themselves in need of explanation. Since now the history of physiology shows that all scientific explanations in these domains have been won through the leading back of organic phenomena to physical and chemical relations, it might seem to be warranted to apprehend the organism all through in analogy with a

²⁵Cf. YVES DELAGE: *Comparative Anatomy and the foundation of Morphology*. (Congress of Arts and Sciences. 1904).

²⁶CUVIER: *Histoire des progrès des sciences naturelles*. Bruxelles. 1838. II, p. 455.

mechanism. But there always remains here the difficulty that with a mechanical whole we can know beforehand the elements by whose joining together it comes into being, just as we can see bricks lying heaped up before they are joined into a building. So external a relation does not obtain between the organism and its elements. One can find again, by analysis of the organism and its processes, elements known beforehand, but not explain these processes as the mere product of such elements. As a biological researcher has aptly said: in biology the whole comes forward as given, and we go back from the form of the whole to the elements it shows itself to consist of; but in physics and chemistry we have conversely the elements given, and the task is to understand their combination into the whole — the atom is not given, but is constructed from the material processes that can be exhibited.²⁷ — Against the analogy with mechanism there has often been set the analogy with human production, now in a more external way, by using the comparison with a master-builder's relation to the building he erects, now more spiritedly, by using [107] the comparison with an architect of genius and the plan that comes into being in his creative imagination. But in both cases one then distinguishes between creator and work, and here the analogy fails. — KANT's result in his investigation of this problem stands fast to this very day, when he declares: "Genau zu reden, hat die Organisation der Natur nichts Analogisches mit irgend einer Causalität, die wir kennen" (Kritik der Urtheilskraft § 65). This does not now exclude that analogy's spurring effect can make itself felt here too in research. Both the analogies adduced can, namely, be applied heuristically, and are in fact applied by different researchers, in that now the question presents itself what physical and chemical conditions an organic process presupposes, now it is asked how far an organism is so equipped that it can maintain its subsistence under determinate conditions, especially when these are subject to change. The future must show whether this position of biology's is the final one.

i. In the question of analogy's significance in the scientific sense there makes itself felt an interesting opposition between two excellent French epistemologists. ÉMILE MEYERSON regards, as we have earlier seen, analogy as the last form of scientific thinking about the subject-matters before us. On his conception, man's rational need finds, by the carrying through of scientifically analysed analogies, the satisfaction that is possible at

²⁷JOHAN HJORT: *The Unity of Science*, p. 166.

all. But in the irreducible qualities there always remain irrational elements.²⁸ Against this conception LÉON BRUNSCHVIG objected that it lays a false ideal of cognition at the base. If true science were to be the carrying through of absolute identity as cognition's [108] highest form, the attainment of this ideal would be one with intellectual suicide. As soon as one gives up this ideal, the supposed irrationality falls away, since it merely marks the opposite of absolute identity. True science seeks points of view which make it possible to find equivalence between complex subject-matters and their elements; but it cannot be set up as its ideal to falsify the qualities before us by transforming them into pure quantities.²⁹

For my own part I have already, in a lecture I gave at Oxford in 1904, printed in *Mind* 1905 and in my *Mindre Arbejder* (second series), professed the point of view of analogy, and in *Den menneskelige Tanke* (p. 181–201 and 309–318) I have taken up this consideration again. The clash between identity and analogy runs through the whole history of philosophy. It has cost, and it costs, resignation to see an analogy carried through as far as possible as the highest that can be reached, and as the only way in which the different roads of cognition can meet, and in which the relation between the different domains of cognition can be determined. But since analogy is only likeness of relation, not identity, there always remains an irrational element — the more so as it cannot be known beforehand whether the analogies now at our disposal can be carried through in the case of new subject-matters, or whether it is possible to find new analogies when unforeseen subject-matters demand them. Beyond exact analogies it would be possible to get only if it could be shown that the inferences, in analogy with which we apprehend the qualitative and temporal differences, could be reversed — so that one could be sure that no other premisses than those on which our sciences build could lead to results [109] agreeing with the observations.

19. Natural Science and the Human Sciences

In the elucidation of the relation between formal and real science, natural science has in the foregoing been used as the example of real science, because it is incontestably

²⁸ *De l'Explication dans les Sciences*. I, p. 217–224; II, p. 291–295 and *Bulletin de la Société française de Philosophie*. 24. Febr. 1921, p. 59.

²⁹ *Bulletin*. 1921, p. 59 f. *L'Expérience humaine*, p. 581.

the most perfect real science. The human sciences (philosophy, the science of language, historical research) work constantly with qualities and successions which it has not been possible to lead back to relations as clear as those which natural science has for an essential part succeeded, by the way of exact analogies, in exhibiting for its own subject-matters. But then it is a matter of investigating the relation between natural science and the human sciences, especially with regard to whether this relation is a relation of identity or a relation of analogy.

One might ask whether there is at bottom any science other than the human sciences. What we call natural science is after all the fruit of spiritual work, and its methods, as well as the results reached by the help of those methods, must be determined by the essence of spiritual work. Can natural science ever become anything other and more than a great attempt to apprehend existence as spiritual work is able to? Natural science is to that extent only a peculiar form of human science, and the difference is only that the human sciences, besides investigating the spiritual work which leads to science, also investigate other forms of spiritual work — for instance those which lead to art, morality and religion. Of special interest in this connection, however, is that epistemology itself is undoubtedly [110] a human science, whose subject-matter is that capacity for work, that energy, which lays itself open to view in the fact that our representations undergo changes leading at once to greater connection among them and to more exact determination of each of them. The epistemological way of regarding things gives us a unifying point of view for all science — perhaps the only one there is. Far from covering over the differences between the sciences, such a unifying point of view precisely sharpens the eye for those differences. The different ways of working, and the different success with which work is done in the different fields of work, gain precisely in interest from such a point of view. And it now also becomes more easily possible to investigate the relation between scientific work and spiritual work performed in other domains of human mental life than the intellectual. Here too the unifying point of view will let the peculiar differences come forward with greater clarity.

But even if a consideration such as the one now indicated should be the fundamental and the definitive one, the matter presents itself in many people's opinion otherwise when we turn our attention to the different fields of intellectual work. When it now shows itself that the work of natural science — whose field is the so-called material

phenomena, among which is reckoned everything that comes forward for us in the form of space — proceeds by more perfect methods and is attended by greater success than the work of the human sciences, which does not get beyond differences of quality and of time, would it not then be reasonable to make all intellectual work into work of natural science — to let the more perfect knowledge elucidate the less perfect knowledge and take it up into itself, by apprehending all so-called spiritual subject-matters as members in the great continuous connection between all so-called material subject-matters which natural science more and more discloses to us? Everyone concedes (however he may otherwise apprehend the relation between “soul and body”) that spiritual subject-matters, whatever constitution they may have, hang together with certain physiological processes; and since these last decidedly fall within natural science’s domain, would it not then be most correct to take one’s starting-point here and regard the natural-scientific apprehension as the only valid one, and then apprehend the human sciences (in so far as they can be called science at all) as a part of natural science? [111]

When I attempt to answer this question, I must first refer to what I have attempted to show in earlier sections of this treatise: that all real science builds on analogy with logical and mathematical thinking, so that the question finally becomes one of that thinking’s validity. Are inferences and equations spiritual or material? If they must be said to be forms of, and results of, intellectual work, it must be asked who draws the inferences and grounds the equations. We thereby stand before the relation between epistemology and psychology, and must dwell a little on it before we go over to the relation between psychology and physiology, which more immediately concerns our problem.

a. As regards the first relation, I must refer to my treatise *Totalitet som Kategori* (p. 4–18; 46–48), where I have sought to show that the circumstance that psychology’s possibility is one of epistemology’s subject-matters does not abolish psychology’s independence, any more than physics’ independence is abolished by its also being such a subject-matter. The epistemological subject which is presupposed in all inferences and equations is now one of psychology’s most interesting subject-matters. By the help of the history of science, psychology seeks to show how this subject has developed from being merely observing and describing to so high a stage that it has been possible to say that it is this subject that prescribes to nature its laws. It is in constant development; at each time it is at a determinate stage of that development. It does not exist in and for itself; that [112]

one can assume only by a leap from epistemology to metaphysics. And at every one of its stages of development it sets psychology the task of exhibiting how its development has been possible at all — and that for the good reason that at every stage it presents specifically psychological, and not merely purely logical and mathematical, properties. It is the principles of spiritual growth that psychology here applies to the subject of cognition which all science presupposes. There are epistemologists who would most gladly slip everything psychological off themselves; they are hit by the irony with which KIERKEGAARD speaks of the speculative philosophers of his time. The Marburg school, so meritorious in other respects, is hit by such irony. Naturally psychology cannot build on other principles than those acknowledged by epistemology; but that it has in common with — epistemology itself, which naturally cannot build on other principles than such as it can acknowledge when it one day gets finished. (Cf. above, § 7.)

b. As regards the relation between psychology and physiology, I can also refer to earlier utterances — partly in my *Psykologi* and my *Psykologiske Undersøgelser*, partly and especially to my university programme (*Filosofiske Problemer*) (1902), whose content is taken up in abbreviated form in *Den menneskelige Tanke* (1910) (p. 22–34; 329–338). I [113] shall in this connection dwell only on a few points. An observing (respectively experimenting) and describing psychology is the presupposition for finding the physiological connection which in some people's opinion³⁰ is to give us the only scientific explanation of mental life. In order to “explain” something, I must first know what it is that is to be explained; otherwise I cannot know whether the right explanation has been found. So-called psychophysiology therefore always presupposes psychology proper. The programme of leading all psychology back to physiology can be carried through only when empirical-experimental psychology has first done its work and thereby made precise the phenomena that are to be explained. Psychology has great difficulties to carry through in its own domain, and it is of no use to proceed precipitately. The discontinuity of mental phenomena, their qualitative differences and their constant changing, as well as their manifold and complicated interweavings, constantly set new tasks for descriptive, genetic and analysing psychology.

If we now look at the physiological “explanations” which are to impress the psy-

³⁰In this direction goes, within our own literature, the acute treatise of JØRGEN FR. JØRGENSEN: *Problemet om Forholdet mellem Sjæl og Legeme*. (The journal “Vor Tid”. 1913.)

chologist by their exactness, it shows itself that they have been won by the help of analogies drawn from direct psychological observations. They are attempts to construct a physiological mechanics answering more or less to what psychological observation shows. It is a psychological fact that the more frequently two representations accompany one another, the more easily the transition later takes place from the one to the other. This is to find its explanation in the “path-forming” of nerve-cells and nerve-fibres. But in reality one knows no more of what such nerve path-forming properly is than what lies in the fact (which is both a psychological and a physiological fact) that a function [114] goes on more easily by repetition. One has got a mechanical image; that is the whole of it. And such an image might very well have been applied directly in psychology itself, since all psychological expressions show themselves etymologically to be fetched from the outer world.

One of the first and most remarkable attempts to apprehend psychology as physiological mechanics was made by THOMAS HOBBS, who will maintain that sensation and thinking are motion in space and nothing else. But he then stops involuntarily at the fact that motion can be apprehended (become phenomenon), and that is not explained by the motion itself. (See the appendix to this section.) In RICHARD AVENARIUS’ acute and instructive work *Kritik der reinen Erfahrung* the attempt is made to carry through the conception that a scientific apprehension of experience and thinking is got only by regarding these functions as processes in the central nervous system. A whole series of such processes is then depicted; but when it is asked what they signify, it shows itself that they are built on analogy with directly observed mental phenomena. The so-called “objective vital series”, which is to be regarded as the independent variable in relation to the “subjective vital series”, is in reality dependent on the latter. (Cf. my characterization of AVENARIUS in *Moderne Filosofer*.) While HOBBS does in flashes come to intimate the subjective presuppositions of physiological psychology, and while AVENARIUS expressly acknowledges that there are two “vital series” standing in a functional relation to one another, BECHTEREW in his *Objektive Psychologie* (1913) has attempted to write a psychology avoiding all psychological designations. But the consequence is that the reader must constantly guess why just these determinate physiological processes are described in [115] this determinate connection. Thus great weight is laid on “concentration-reflexes”, in opposition to sporadically isolated reflexes, and the reader then guesses that it is attention

that is being thought of. The expression “concentration” can very well be used in a purely psychological sense, of our gathering of the mind in a determinate direction, just as we saw before that one speaks of path-forming as a good image for a practised course of thought. The place given to the “concentration-reflexes” is due to an analogy with mental concentrations, such as descriptive psychology can present them. BECHTEREW himself concedes that to the difference which observing psychology draws between the unconscious and the conscious, the involuntary and the voluntary, he can find nothing physiologically corresponding; and therefore, he says, this difference has no significance for objective psychology. If objective psychology must give up before differences of the kind stated, then it has not much significance. I think now that BECHTEREW gives up too early. Even if one does not think that physiological psychology is the only scientific one, it can and must all the same be set as an inescapable task (whether it can be solved or not) to find physiological correlates to those differences so significant in the psychological respect.

Physiological processes are directly measurable in a degree in which psychical processes are not. When we set the two kinds of process together, we stand again before the relation between a quantity-series and a quality-series. Here too identity and analogy must not be confused. The members of one of the two series can be used as symptoms of what goes on in the other. If one cannot connect any meaning with a thought's being the
 [116] product of the reciprocal action of material elements, or with a thought's bringing forth an alteration in the body, then one must stop at a relation of analogy³¹ which may perhaps one day become a relation of proportion. The conception we come to by keeping to what can be observed is that there are in any case some beings which have both psychical and material properties answering to one another, without the one group of properties being derivable from the other. It is a riddle that here remains — an example of the fact that where we must stop at an analogy, we stop at a question. Such a relation may for that matter occur in purely psychological form. There may be in a man's character properties that cannot be derived from one another, and which we therefore do not understand, because we do not know the character itself deeply enough to see how the two different

³¹For the rest, the popular theory of reciprocal action would also build on analogy, since it presupposes that the relation between soul and body is analogous to the relation between two bodies (two billiard balls, e.g., or — when it is to be more “scientific” — two groups of atoms). DESCARTES lets body and soul “collide” with one another, and LOTZE has nothing against this theory.

properties hang together in their inmost ground. As regards the relation between the spiritual and the material, we are so placed that we can now infer from mental states to states in the organism, now conversely. SPINOZA, the founder of the conception to which I here most nearly attach myself, does indeed already go both ways.³²

It is finally the relation between the concepts of personality and organism that we here come to stand before. What KANT said of the organism, that at bottom no analogy to it can be found, holds for the personality too. So intimate a connection as that between [117] the elements of mental life we know no parallel to. The more interesting is it that those two concepts are mutually analogues.

Appendix to § 19. HOBBS apprehended everything, sensation and thinking too, as motion in space and *nothing else*. But at one place especially it dawns on him that the matter is after all not so simple. It is finally through sensation that phenomena — that is, motions — come into being for us, and that science becomes possible. And when it is then asked after the causes of sensation itself, we cannot take our starting-point from any other phenomenon than precisely that one (sensation) itself (*ad causarum sessionis investigationem ab alio phænomeno præter eam ipsam initium sumi non posse*. De Corpore XXV, 1). It is on this account that HOBBS declares that of all phenomena the most remarkable is the very fact that anything can become a phenomenon (*id ipsum τὸ φαίνεσθαι*), *admirabilissimum*.

I understand this remarkable passage in the following way. That anything can become a phenomenon HOBBS finds so remarkable because a motion cannot become a phenomenon for another motion (that is, be *known* by it). The one motion can continue the other, hamper it or stop it. Sensation must then be different from *mere* motion, though it is *also* motion. Now since it is through sensation that we know everything, it must also be through sensation itself that we know sensation. But to sense one's own sensation is to remember; and there is accordingly, HOBBS maintains, memory and comparison bound up with all sensation. (*Sentire se sensisse, memoria est. — Sessioni necessario adhæret memoria aliquæ, quæ priora cum posterioribus comparari et alterum ab altero distingui potest*. De Corpore XXV, 1; 5.) It is in consequence of this conception of sensation that [118] he sets up his famous proposition that in all sensation there is contained a manifold (*phantasmatum varietas*), and that always to sense the same and not to sense at all come

³² Spinoza's *Ethica*. Analyse og Karakteristik, p. 48 f.; 87 f.

to the same thing (*sentire semper idem et non sentire ad idem recidunt*) (5). From this purely psychological observation he then infers that the motion to which sensation is bound must be capable of being held fast and recalled, and that there must be conditions for this in the sense-organ. More clearly the difference between the subject of cognition and the motion which according to HOBBS is the only thing objectively existing cannot be expressed. But HOBBS was too much of a dogmatist for this flash of thought to make him give up his metaphysics of motion. It went with him as it later went with SCHELLING, who (as I have shown in *Den nyere Filosofis Historie*³ III, p. 158) had an analogous flash of thought, in setting up very clearly the problem: how can there exist not only nature, but also a science of nature? This flash hindered SCHELLING as little from going further in his Romantic philosophy of nature as HOBBS' flash hindered him from going further in his mechanistic philosophy of nature.

In FRITHIOF BRANDT's excellent monograph *Den mekaniske Naturopfattelse hos Thomas Hobbes* (1921), in which light is thrown on so many hitherto overlooked or misunderstood points in HOBBS' philosophy, the passage adduced above is discussed at length (p. 365–373). The author seeks to show that there is in reality no inconsistency here in HOBBS. But no account is taken of the proposition which in my eyes is decisive, that the understanding of sensation must be won from sensation itself, nor of the consequent [119] circumstance that from the purely psychological understanding of sensation's nature one must *infer* to the constitution of the sense-organ and to what must go on in it.

My interpretation of the utterance of HOBBS in question as a flash of thought leading beyond his absolute metaphysics of motion is confirmed by his statement (*De Corpore* VI, 7) that one can come to ethics and political science without going the road of natural science, but merely by building on the psychological experiences of need and impulse which everyone can make directly in himself (*per uniuscujusque proprium animum examinantis experientiam*). Not only need and impulse, but sense-sensation too, we know directly by this road.

20. Ethics and Theoretical Science

When ethics is put in a certain opposition to theoretical science, this does not rest on one's having the right within ethics to think and infer in another way than within all other science. There is no special ethical thinking as regards grounding and the conducting

of proof. But ethics has presuppositions which other sciences need not set up. Every thinking, however theoretical, rests on the presupposition that truth has value, that it is valuable to seek after it. To such a degree is this presupposition necessary that one does not even consider it necessary to name it. When in the present a philosophical tendency, whose most considerable representative is HEINRICH RICKERT, has all the same laid weight on this presupposition,³³ the warrant lies in this, that the problem of value is thereby drawn out of the isolation which can easily follow when it is seen only in its significance for aesthetics, ethics (economics included) and the philosophy of religion. When such isolation is averted, there comes forward a fundamental connection in human spiritual life, a connection which lies deeper than the difference between the different [120] problems, but which at the same time grounds the difference that there is. When science is preferred to artistic or religious striving, then it is value that stands against value, and a clash takes place between different spiritual needs. The problems come into being by each of these spiritual needs setting thoughts in motion. It then depends on whether these thoughts can be brought into agreement with one another. But here it is that the great significance of the value of truth, all science's tacit presupposition, comes forward, precisely in relation to all other values. The thoughts which the different values set in motion must — for the values' own sake — be subjected to a purely theoretical testing, a testing whose sole purpose is to find the thoughts' grounding and their mutual relation: a testing, that is, from the value of truth as the ruling one. Of all the problems, therefore, the problem of cognition takes precedence. On the result to which the discussion of the problem of cognition leads depends the validity that can be ascribed to thoughts set in motion by other motives than the purely intellectual. In the treatment of the problem of cognition it is presupposed that we have no other interest than that of finding the truth, however this may be constituted. It then becomes the question how other interests that stir in us can be united with this interest. In my book *Den menneskelige Tanke* (see especially p. 22) I have therefore proceeded by presupposing that an intellectual interest has developed, a joy in thought-work itself, which leads to thought's being able to develop according to its own laws and according to the demands of the subject-matters. The other interests, for which thought-work is merely a means and not an end in and for itself, will finally find themselves best served by the intellectual interest's ruling for a [121]

³³*Den nyere Filosofis Historie*³ IV, p. 219.

time, and by a theoretical revision being undertaken from time to time of the thoughts they have especially called to life.

In the ethical domain it cannot now be avoided that interests other than the purely intellectual make themselves felt. All ethical thinking presupposes a fundamental value and aims at finding means and ways to maintain it in the judgments that are passed and the social orderings that are formed. In an apt way HENRI POINCARÉ has in his treatise *La Morale et la Science*³⁴ expressed this by saying that ethical thinking has a premise which is imperative, since it springs from a need demanding satisfaction, while the other premisses are indicative, since they state the means and ways by which the satisfaction can under the given conditions be reached. Since there is an imperative premise, the conclusion must necessarily also be imperative. The consistency that can here be in question becomes a consistency of the relation between means and ends. If the grounding of the need or value which gives rise to the imperative is asked for, the answer must — since value can be measured only by value — be fetched from the fundamental value which the special value presupposes. And if the grounding of this fundamental value is asked for in turn, we must keep to the fact that every determination of value presupposes a whole. Values do not float in the air, but are the expression of a whole's need for subsistence. It may be a single moment, a period of life, a single interest, that one makes into one's all in all; it may be one's own personality, or the class and stock it belongs to, or finally the subsistence and development of the whole

[122] human race, that constitutes the last presupposition of imperative ethics. The dispute of theories rests finally on the fact that different wholes can be laid at the base, and that thereby the fundamental values too become different.³⁵ An attempt at scientific ethics can here proceed only by taking the starting-point from a determinate whole, drawing all inferences from the conditions of its subsistence and development, and thereafter seeking to show how the other wholes are to be valued in accordance with the results so won. In this last investigation there is special ground for analogy. Even if the other wholes, in virtue of the starting-point, must be treated as mere means, such treatment becomes purposeful only when there is an understanding of their nature. And if they are more than means — if a relation of solidarity or even of sympathy follows from the given

³⁴*Dernières Pensées*. 1913, p. 229.

³⁵The connection between the concepts of totality and value I have exhibited especially in *Den menneskelige Tanke* 145–146; 151, and in *Totalitet som Kategori*, Kap. V.

starting-point — a still more deeply going understanding of them is required; fortunately in such cases the capacity for understanding can grow with the need. Between peculiar wholes there is a relation of analogy, not a relation of identity. Ethical thinking's validity (from the given starting-point) rests on the strictness with which the analogies are carried through.

One might think that the most comprehensive whole must in and for itself furnish the right starting-point for ethical valuation, so that its subsistence became the self-evident fundamental value. But besides extent, inwardness and strength also come into consideration with the concept of value. Within a less comprehensive whole there may be an inwardness and a strength in the union which do not so easily arise within a greater whole; and the greater whole's value will for an essential part rest on the independence it is able to acknowledge and to further in the smaller wholes it comprehends. [123]

By the analogies between the different wholes it becomes intelligible that there are certain ethical concepts which are formed, and must be formed, at all standpoints, because they express the formal relation between a whole, regarded as fundamental value, and the conditions of its subsistence and development. Such formal ethical concepts are right and duty. The single whole is to maintain itself against other wholes and against outer conditions generally; that is its right, which follows from its position as fundamental value. But its task is besides to guard the means to its subsistence, and also to work for the subsistence of those wholes which are means for it, or to which it stands in a relation of solidarity or of sympathy. Under the concept of duty belongs prudence, the understanding of the nature of the means and of the necessity of procuring them. Especially with regard to the significance and application of these formal concepts, the one ethical standpoint will be able to learn from the other. It is impressed on us, as regards the duty of prudence, by the parable of the unjust steward (Luke XVI), which shows that the children of darkness often have at their disposal, within their own sphere, a prudence which is wanting in the children of light in their work. It has been said in our own day by an experienced statesman that the task of clever men often consists in remedying the misfortunes good men have caused. — LEIBNIZ has defined the concept of justice in such a way that right and duty, humanity and prudence are united in it: *caritas sapientis!*

Not only between different ethical standpoints among themselves can analogy make

[124] itself felt in a fruitful way, but also between ethical consciousness and scientific consciousness. The scientific consciousness, which seeks clarity and understanding for their own sake, can become a model (a Paradigm, to use Aristotle's expression) for ethical consciousness, both as regards disposition and as regards consistency. The same can be said of the artistic consciousness in its striving to achieve the perfect. Nothing, says MICHEL ANGELO, makes the soul so pure and noble as striving to accomplish a perfect work. In all spiritual domains it is a matter of uniting unity and closedness with the free unfolding of a fullness of force. Justice (in the sense of the word adduced) may be called the ethical synthesis, as truth is the intellectual synthesis. (Den menneskelige Tanke, p. 365–367.) In the treatise on morality and science adduced above, HENRI POINCARÉ says: "Intellectual habits have a moral resonance. When one accustoms oneself to disregard particulars and accidents, because they are of no importance for the understanding, there will arise a natural inclination to subordinate particular interests to general ones altogether".

21. World-view and Science

It is a true word of WUNDT's that it is easier to reject all metaphysics than to act in accordance with that rejection. By many hidden ways combinations and constructions steal into the background of consciousness, and now and then they even press forward into the foreground. Again and again a critical investigation of such constructions will be necessary. And first and foremost it is of importance that one become clear about the constitution of the problem one here stands before. As I have attempted to show in another place (Den menneskelige Tanke, p. 306 f.), there is an analogy between the problem of existence and the problem of cognition. Cognition becomes, as regards its validity, a problem through the fact that human thought is indeed a fact, an actually given subject-matter, but that it is not the only subject-matter and yet is to hold both itself and all other subject-matters. It is to develop according to its own laws and yet to build a world in which all subject-matters, itself included, can find a place answering to their peculiarity. Thought is to rule and is yet an element in a world greater than itself. The world is great and thought is small, however energetically it stirs. The problem of existence — that is, the question of the possibility of a scientific world-view — is likewise due to a relation between part and whole. But here the question is whether any single

subject-matter, any part of existence, can stand as the type for the whole of existence, as that primal phenomenon which gives the characterization of the whole of existence in brief. All historically given world-views, religious or speculative, are crystallizations, involuntary or constructed with conscious reflection, about a primal phenomenon. In the series of these crystallizations one can read off which subject-matters have at different times drawn interest to themselves and become unconscious or conscious central points in men's minds. — It is clear that in the coming into being of a world-view the thinking is by analogy, ἐν παραδείγματι as Aristotle says. One subject-matter is used as a model for all subject-matters.

We have here operated without more ado with the concepts of part and whole. But have we the right to start out from the assumption that existence constitutes a whole? — BERTRAND RUSSELL has declared that the representation of a universe as a whole is a remnant of pre-Copernican astronomy which not even KANT's criticism has been able to get rid of.³⁶ It is doubtless also so that it is the sensible picture of the world that determines the representations in this domain, even though science has long since given [126] up its absolute validity. But there is all the same a deeper foundation for the tendency that here makes itself felt. The fundamental law of all categories, as of the life of consciousness generally, is the law of synthesis. There works here, to use KANT's expression, a hidden art in man's interior which weaves sense-sensations into a sense-image, sense-images into concepts and laws, and finally, by the help of the lawlike connection, a great image of the whole, more or less carried through and elaborated. Criticism always comes afterwards. And the most important critical objection against an attempt to fix such images of the whole is that it would be the death of thought. Since all thinking is finally a gathering-together, there is presupposed in all thought-work a manifold, a discontinuity, which is to be overcome. And with a complete and concluded satisfaction of the need for wholeness, that very "hidden art" would die a straw death. It would no longer do what in its finest and noblest forms it has been able to do: shape great ideals, set goals pointing beyond what has hitherto been won. Discontinuity must have a place among the fundamental categories, to remind us that there is always work to be done. Every whole that is found can mark only a provisional station. Our cognition works constantly on a scale whose highest step would be the unattainable absolute identity-series, and

³⁶*Mysticism and Logic*, p. 987.

whose lowest step would be the undemonstrable chaotic difference-series.

In the most pronounced cosmological systems — PARMENIDES', SPINOZA'S, HEGEL'S — identity has been the last and highest point of view. The ideal was to apprehend everything as in the last instance one. It was at bottom an oriental ideal that here came to [127] rule in European philosophizing. A turning-point came with LEIBNIZ, for whom identity marks a methodological principle in our grounding and proving course of thought and can acquire cosmological significance only by the way of analogy. LEIBNIZ maintains precisely, with great emphasis, that all cosmology (metaphysics), and therewith all theology too, builds on analogy — finally on analogy with the human spirit and the harmony and wholeness that can be found in it, or that it at any rate strives after. Still more searchingly KANT sought to draw the boundary between the science that works with the principle of identity and the faith that works with analogies. At the beginning of the nineteenth century new attempts were made at a cosmological doctrine of identity. But it seems as though the point of view of analogy is winning more and more adherence, even among thinkers of a predominantly speculative tendency.³⁷ The figurative character of the representations which are formed, involuntarily or with full consciousness, at thought's boundary and beyond it, is more and more making its way. This figurative character indicates that a choice has been made between the mutually analogous series — the formal (logical-quantitative) and the real (qualitative-successive) series, and within the real series the natural-scientific and the human-scientific. But the analogy between these series is and remains science's last result. A scientific conclusion would require the identity of the series, not mere analogy. It simply is so that our exact concept of reality comes into being through the formal series' making the real series intelligible, and that the formal series are themselves a side of reality — the side we become aware of as soon as we try to orient ourselves in existence.

[128] How one takes one's stand at the very boundary of scientific thinking will depend in part on which epistemology one adheres to. There is here a somewhat different situation for pragmatism and for criticism. Both conceptions maintain that the thoughts which can be shaped when we have reached that boundary can only be analogies. But pragmatism, which on the whole is inclined to ascribe validity to all thoughts springing from human need, and which is not interested in a critical investigation of our fundamental concepts,

³⁷Cf. e.g. BOSANQUET: *The meeting of extremes in contemporary philosophy* (1921), p. 17.

will also have no interest in investigating how far one can still trace, in the limiting concepts which are human cognition's last word, the fundamental forms characteristic of it. Criticism, on the other hand, will be interested in finding again the structure of the human spirit beyond the domain in which scientific concepts can be formed; in the very kind of the figurative representations that are still possible beyond that domain's boundaries, it will seek for peculiarities of human spiritual life. It lays weight on thought's homologies, on the kinship which a common structure can condition, not merely on the analogies that arise from thought's having to be adapted to new conditions.

Cosmology cannot become science. It stands between science and art. It forms the transition to poetry and religion, the domains in which the need to find concrete expressions for the experiences of life shows itself freely and involuntarily. Even there philosophy will seek not only analogy but also homology, since the nature of the human spirit will lay itself open to view there too.

Chapter V

Poetic and Religious Symbolism

22. Out of that merging into involuntary analogies which is peculiar to human spiritual [129] life at the most primitive stage we know, there has developed, through many transitional forms, the scientific consciousness which demands identity as the strict presupposition of every transition of thought, and which, where this presupposition cannot be exhibited, works with rational analogies. Science's development is a member in the great spiritual division of labour that comes with advancing culture. But when one element is released from the primitive connection in which all spiritual forces are joined undifferentiated, other elements are released too, or at any rate get other conditions to work under. Alongside, and more or less under the influence of, the independence of intellectual culture, poetry and religion too have in greater or lesser degree released themselves from the original boundness. In opposition to the rational analogies there then gradually come forward the emotional analogies characteristic of poetry and religion.

The immediate absorption, or rather enthrallment, by a subject-matter can give rise to an equally immediate need to shape it in an image or in words, and thereby to make known its value for oneself and for others. We live or feel ourselves into the subject-matter, strive to follow it in its fate and to get a share in its properties. The individual and immediately intuitable, which no science can exhaust in its concepts or in its laws, poetry gives us in its images, sprung from lived experience and living-into — and perhaps even with so complete an individualization as the subject-matters given in experience did not have. And what thought can express in pale forms — the opposition between the low and the high, the bitter and the sweet, the fleeting and the abiding — poetry, [130]

especially as drama, leads us immediately into. Whether what is given us in this way is the rule or the exception, poetry cannot itself decide. Every example is a symbol, but not every symbol is an example. (Cf. above, § 10.) Poetry can set problems only in so far as the very fact that a single, unique, individual subject-matter belongs to existence can serve to characterize existence, or can at least give occasion for serious reflection. But often the emotion awakened by the subject-matter will be so strong that no single symbol satisfies, and a whole series of symbols is developed. It is a series-formation of another kind than the intellectual one. (See § 5.) Not the inner, logical relation between the members of the series, but the emotion underlying all the members, here conditions the continuity and binds the members together into a series.

It marks, as SCHILLER already saw, an advance of culture when there can be joy in the image, apart from whether the subject-matter is in our possession or not, and apart from whether it has any significance as a sign or as a means. So long as “participation” (“realism of the whole”) rules, this joy cannot arise. But even when the image is freed from practical need of one kind or another, the joy in it does not thereby become of a purely intellectual kind. It is a displacement that has taken place, and with the displacement there follows an ennobling; the interest has moved from the subject-matter’s reality to the image’s ideality, and has thereby itself undergone a metamorphosis.

A distinction must here be drawn between the mind that gives birth to the image and the mind that is affected by the image once it has come into the world.

[131] In the poet, feeling is the primary thing. The poetic images or similitudes proceed from a feeling of the subject-matter’s value, which awakens a need to dwell on it and to see it from all sides. Perhaps the mind gets air only through a whole series of similitudes. These then present analogies among themselves, but these analogies are due to the inwardness with which the poet’s mind holds fast to the subject-matter and its value. Thus there is analogy between Jesus’ parables among themselves, because they are borne by one and the same fundamental mood — that awakened by the idea of the kingdom of heaven.

The reader or hearer is here otherwise placed than the poet himself. The image or the similitude shows him, perhaps with striking clarity and in an expression of the whole, the relation that is in question. And in relation to this intellectual effect, the feeling becomes derived. The poet need not have set himself the express purpose of bringing forth this

effect. In the moment of inspiration he in any case does not. He sings as the bird sings.

In his interesting treatise *Metafor och Fiktion. Ett bidrag til Poetikens Teori* (Lund 1922), Dr. NYMAN relies on the statements of experimental subjects to whom different poetic images were presented. Here it follows of itself that the feeling becomes secondary, and that the image's effect in calling forth an intellectual bird's-eye view comes to stand in the first rank. But for a full aesthetic appreciation of the significance of images and similitudes, one must ask both after the origin and after the effect. And the poet one cannot make into an experimental subject, so as by that road to find out how the image comes into being. He must be spied on. And JULIUS LANGE has shown himself to have spied well, when in his definition of artistic value he set as the first member "the value a subject-matter has for the artist". In return, he laid too little weight on the ways and means by which that value could be called forth for hearer or reader. (Cf. above, § 4 b.) [132]

All poetry stands in connection with life and its fates, and it may therefore become difficult to distinguish between poetry and religion, to decide where the one ends and the other begins, or conversely. Religion too speaks in images and symbols. The distinction that can be drawn rests essentially on this, that the shaping of the images *as* images plays a greater part for poetry than for religion. The fact is, no doubt, that the underlying experiences are here deeper and more strongly enthralling, and address themselves, with the demands that spring from them, more to the will than to the imagination. The need to live immediately in the symbols, and thereby to get a share in the experiences from which the symbols sprang when they were formed at first hand, is characteristically religious; and on that rests the great part tradition often plays in religion. It is not only because tradition in and for itself constantly exercises its power over men's minds that dogmas and forms of worship maintain themselves so long. It is also because there the individual stands before symbols in whose coming into being he knows he has had no part, so that image and thing come to stand as one, and immediate living-into becomes more easily possible. As little as the individual is conscious of having had a part in the symbol's formation, so little does he distinguish between interpretation and lived experience, between image and reality. The poet complains that "life hovers between vision and possession"; but this difference does not exist at pronouncedly religious stages. There will here in any case be a point where vision and possession coincide. On this it rests that religion bears the stamp of the primitive standpoint of participation, and of its

involuntary analogies, longer than science and poetry do.

[133] The divorce between lived experience and interpretation, between possession and vision, scarcely goes on without pain. The religious problem, as it presents itself in our own day, concerns essentially the question whether this divorce can be carried through without harm to the energy and concentration of spiritual life.

For in lived experience we are receptive; life streams in over us with its mighty waves. In interpretation we are more self-active; it is a damming in and a limiting, a self-defence against the strong influences. The old Frisians said that God made the sea, but the Frisians the dykes. (*Deus mare, Frisones littora fecerunt.*) The damming in may be necessary to preserve the fertility of the soil on which life is to be lived. But here individual forces intervene, and there may even come a tension between lived experience and interpretation. So long as tradition gives us both at once — or rather, so long as it is not seen that tradition itself contains an interpretation — no properly religious problem arises. There can at most arise the question how what has been handed down is to be appropriated and understood.

But the real problem religion has in common with cosmology. In every symbol a part serves to elucidate the whole. In the religious symbols too, a part or side of existence is to elucidate, indeed to reveal, the whole of existence's inmost essence. Men cling, in all forms of religion or religiosity, to a single subject-matter in the confidence that it marks a breaking through of the highest thing of value. The choice stands between this determinate interpretation and the possibility of a chaos of valuelessness and meaninglessness. In PASCAL one can study this Either—Or particularly, which forms an analogy to the logical opposition between a chaotic difference-series and a partial or absolute identity-series. There was for PASCAL only one single gleam of light in the darkness of existence. The profound JACOB BÖHME too, deeply seized by the great riddles, halted for a while at this Either—Or. But he seized the way out of facing the oppositions and apprehending existence as a great battlefield where the strife stood between light and darkness, purity and lust, love and hate; and he found here the foundation for a bold ethical conduct of life, while PASCAL's dilemma led him to strict and solitary asceticism as the only way out.

When a part is to stand as the expression of the whole, there will involuntarily make itself felt a striving to prove that the part in question really is typical — perhaps even

that all other subject-matters that experience presents can be derived from it. Thereby religion becomes theology and speculation, and sophistries lie near. As we have seen, an example is always a symbol, but a symbol not always an example. What becomes a symbol may be a *unicum*. The religious poet says of his God that he “saves the trembling dove”. But are there not many poor doves that are not saved? And if they were all saved, how would it then go with the goshawks, which are after all subject-matters too? Nor is there any lack of attempts to take the hawks for symbols. — Religious symbolism at its height will regard the subject-matter it originally started out from as derived in relation to the generalization or absolutizing of that subject-matter to which it involuntarily leads. The father-symbol is originally fetched from the fatherly love that can be experienced in human relations. But then the relation is turned round, and the fact that fatherly love can be practised in human relations is derived from God’s being Father (Eph. III, 15). (Cf. above, p. 32.) — Or it is a single historical figure which, by purity of mind and by its love [135] for men, makes vivid the wish to see in it a revelation of the world’s inmost ground. But not only do there stand over against such a figure types of a quite different kind. There also arises the question whether any single figure at all is able to unite in itself all the valuable properties which, through the changing ages and cultures, must be presupposed in one who is to be the constant model for striving men.

STANLEY HALL has in a remarkable and richly contentful book (Jesus Christ in the light of Psychology. 1921) made an attempt to show that the figure of Jesus, when apprehended with historical criticism and with a thorough psychological sense, is and remains the great model for human life, under all life’s fates and in all life’s striving. STANLEY HALL thinks that by psychology’s help he has got beyond the opposition between orthodoxy and criticism, and in particular that he has overcome the difficulty which arises when the model is to give guidance in the face of interests, tasks and experiences which did not, and could not, lie before the historical figure that has been made into the model. The Jesus of the “new age” whom STANLEY HALL would present, and for whose presentation he uses all the resources of the newer psychology, is to unite in himself all great ideals, no matter where they may come from. He is clear that between these ideals, proceeding from different regions of life and history, there will be oppositions so great that a complete harmony will be impossible; but he maintains that it is not possible in any personality at

[136] all.¹ But the question is whether a new image of personality is really brought forth by combining the ideals of many different ages and cultures. And even if it were possible, the new model would all the same become so essentially different from the one that has worked on men's minds through centuries, that continuity with the historical figure which is still to lend its name would be broken.²

Neither in models nor in dogmas can the continuity of human spiritual life be sought, but in the deep need which, in reciprocal relation with life's course, its gifts and its tasks, its heights and its hollows, gives rise to the changing symbols — between which, therefore, analogy but not identity can take place. The analogy makes it possible to learn from models and dogmas other than our own. But identification is impossible, since it conflicts with life's nature and conditions.

The more religion becomes a poetry of life without dogmatic tendency, the less will it assert that the sources of value which it hopes will continue to flow, however much resistance they may meet, are at the same time the sources from which all the subject-matters given in our experience spring. It will be enough for it that a stream of value constantly goes through the world, and it will awaken a striving not only to get a share in what this stream carries with it, but also to clear away the hindrances to its course, so far as it stands in men's power. —

[137] In my *Religionsfilosofi* these thoughts are more closely developed. The purpose of this concluding part of the treatise has only been to intimate that the emotional analogies which make themselves felt in practical and religious symbolism are peculiar counterparts — though in part also opposites — to the rational analogies which are of such great significance in science. Both kinds of analogy stand more and more in opposition to the analogy that lies hidden in primitive "participation".

¹He will come to seem a baffling paradox, until it is understood that personality means a synthesis of elements too manifold and diverse ever to be completely harmonized (p. 238).

²STANLEY HALL's predecessors in religious-psychological research within American literature — especially STARBUCK, COE and LEUBA — strongly emphasized the one-sidedness of the traditional shaping of the image of Jesus. And COE in particular also lays much weight on what is questionable in regarding a single type as alone warranted. See my treatise *Om nogle religionsfilosofiske Arbejder fra den nyeste Tid*. 1903. (Mindre Arbejder. Anden Række).

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